

June 7, 2026 10:00 PM GMT

Pharmaceuticals

The Ozempic cliff: generic risks, opportunities and bottlenecks

The advent of Ozempic (semaglutide) generics in Canada, India and Brazil provides insights into the future of the diabetes market, with consequences for originators (Novo, Lilly, and other branded companies) and for generic companies, CDMOs and API/device suppliers.



Societal Shifts

A Morgan Stanley Research
Key Theme of 2026

We leverage Morgan Stanley Research's global footprint to gain insights into the impact of semaglutide generics across local markets. The arrival of generic versions of Novo's GLP-1 Ozempic (semaglutide) is set to drive the next shift in the diabetes and obesity markets, with the first countries a testing ground for the US and Europe. Following the expiration of market exclusivity, generics of Ozempic have been launched in India and Canada,

and the first generic should launch in Brazil in the coming weeks. We leverage granular data on sales, volume and market share of GLP-1s in India to get the first picture of a post-generic GLP-1 market. We also map the market formation in Brazil and Canada, where we expect to see sharp price erosion, volume expansion and the emergence of a 2-tier market with lower price semaglutide and higher price differentiated GLP-1+ medicines (Eli Lilly's Mounjaro and future innovative drugs).

We believe that insights on generic market formation in the first countries will provide investors with perspectives on how the global diabetes market will continue to evolve. Following the market developments in these countries will provide investors with insights into how the US and European diabetes markets could evolve in the next wave of patent expiries in the early/mid- 2030s. This will be key to assessing the outlook for branded obesity/diabetes drugs, starting with Lilly and Novo, but also for future innovative launches in the category from other originators (Amgen, Roche, AstraZeneca, Pfizer, Structure Therapeutics amongst others). We expect category winners to emerge within the generic industry, with generic companies leveraging global footprint and integrated supply chain more likely to capture the benefit of category volume expansion.

We do a deep dive into the emerging global supply chain for semaglutide generics. This report also includes a deep dive into the formation of the global supply chain for generics to address a key investor debate: as generics accelerate the affordability of GLP-1s and drive greater category penetration in diabetes and obesity, can the generics industry actually supply this greater demand? We believe that China suppliers are scaling enough capacity to provide generic makers with the required semaglutide active pharmaceutical ingredient (API) to manufacture injectable generics (although we believe that scaling semaglutide oral pill will be

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Thibault Bouterin

Equity Analyst

Thibault.Bouterin@morganstanley.com

+44 20 7425-9688

MORGAN STANLEY & CO. LLC

Terence C Flynn, Ph.D.

Equity Analyst

Terence.Flynn@morganstanley.com

+1 212 761-2230

MORGAN STANLEY ASIA LIMITED+

Laurence Tam

Equity Analyst

Laurence.Tam@morganstanley.com

+852 2239-1753

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

Binay Singh

Equity Analyst

Binay.Singh@morganstanley.com

+91 22 6118-1158

MORGAN STANLEY C.T.V.M. S.A.+

Mauricio I Cepeda

Equity Analyst

Mauricio.Cepeda@morganstanley.com

+55 11 3048-6272

MORGAN STANLEY & CO. LLC

Kallum L Titchmarsh

Equity Analyst

Kallum.Titchmarsh@morganstanley.com

+1 212 761-2351

MORGAN STANLEY & CO. INTERNATIONAL PLC+

Sarita Kapila

Equity Analyst

Sarita.Kapila@morganstanley.com

+44 20 7425-1320

MORGAN STANLEY & CO. LLC

Alexander Yevdokimov, Ph.D.

Research Associate

Alexander.Yevdokimov@morganstanley.com

+1 212 761-2167

MORGAN STANLEY C.T.V.M. S.A.+

Artur D Alves

Equity Analyst

Artur.Alves@morganstanley.com

+55 11 3048-4278

MORGAN STANLEY INDIA COMPANY PRIVATE LIMITED+

Dhara P Shah

Research Associate

Dhara.Shah@morganstanley.com

+91 22 6118-2047

MORGAN STANLEY ASIA LIMITED+

Marco Wong

Research Associate

Marco.Wong@morganstanley.com

+852 3963-2021

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

more challenging due to greater API volumes required). A bottleneck could emerge in the fill-and-finish (F&F) part of the injectable semaglutide generic supply chain - echoing the supply challenges faced by Novo and Lilly over 2022-2024. Here, capacity investments over the next few years and the introduction of vial formulations could potentially address the challenge.

We continue to prefer Eli Lilly over Novo Nordisk amongst originators; we are also watching for the emergence of winners in the generic supply chain. We see a challenging outlook for **Novo Nordisk's** semaglutide franchise as we expect Canada to be a bellwether for the US and Europe with pressure from generic competition. While a successful launch of Wegovy Pill could be a partial offset, the scalability challenges specific to Novo's pill will limit geographic expansion. Key new pipeline launches in diabetes (cagrilintide, zenaglutide) are skewed to 2028-2030, creating a risk for Novo's diabetes franchise to struggle between inexpensive semaglutide generics and innovation from Lilly and future branded competitors. We believe **Eli Lilly** will retain a leading position in the higher value branded market given differentiation from its existing portfolio (Zepbound/Mounjaro/tirzepatide demonstrated superiority over semaglutide in type 2 diabetes and obesity), a scalable oral GLP-1 pill (not a peptide) with global reach (Foundary/orforglipron), and upcoming pipeline drugs (next-gen injectables such as triple agonist retatrutide and selective amylin agonist eloralintide). Historically the entry of newer innovative drugs, with differentiated mechanisms has led to market growth in chronic diseases, despite the entry of generic versions of older generation drugs. While it is early to assess which generic companies will benefit from the volume expansion of the GLP-1 category, we expect players with global reach and integrated supply chain to have an advantage (**Dr Reddy's** first to launch in Canada/India), while local players and supply chain participants could benefit from the first wave of generics (**Hypera, Raia Drogasil** in Brazil, **United Lab** and **Medtide** in China).

Exhibit 1: Semaglutide IP and generic situation across geographies

Expected Gx entry	Country	IP situation	Generic situation	Estimated 2025 sema sales
2026	Canada	Expired (4 Jan 2026)	2 generics launched, 7 more lined-up	~\$1.5-2bn
	India	Expired (20 March 2026)	26 Gx launched from 13 companies	~\$50m
	Brazil	Expired (20 March 2026)	1 Gx approved, 13 generics	~\$1-1.3bn
	Turkey	Expiring 2026	SDZ, DRL preparing launch	-
	South Africa	Expiring 2026	Aspen gearing for launch	-
2027	Saudi Arabia	No IP filed	Hikma planning a launch by end 2026	-
	China	Patent expired; Regulatory data protection until April 2027	13+ companies preparing to launch	~ \$1bn
	Mexico	Expiring 2027	SDZ, DRL preparing launch	-
	South Korea	Expiring 2027-2030	Expecting generic launch at IP expiry	~\$300-350m
2027-2030	Baltics, Central and Eastern Europe	Expiring 2027-2030	Expecting generic launch at IP expiry	-
2031	European Union	IP until 2031	1 generic tentatively approved (Apotex)	~ \$4bn
2032	Japan	IP until 2031	Expecting generic launch at IP expiry	~\$1bn
	USA	IP until 2032+	1 generic tentatively approved (Apotex)	~ \$22.5bn

Source: Morgan Stanley Research, IQVIA

MORGAN STANLEY & CO. LLC

Jason Lai

Research Associate

Jason.Lai@morganstanley.com

+1 212 761-4848

PHARMACEUTICALS

Europe

Industry View

In-Line

The Story In Numbers

1 OBESITY — A LARGE AND EXPANDING OPPORTUNITY

People with obesity in 2025

 **1 billion** Worldwide

 **110 million** in USA

☞ Among **110mn** US patients, employer opt-in **~50%**

☞ Medicare opens access to **~20–22mn** patients



2 GENERIC SEMAGLUTIDE GLOBAL MARKET FORMATION: COUNTRY BY COUNTRY

Canada (Blueprint for US/EU)



- **~\$2–2.5bn** GLP-1 market (2025); **~\$1.5–2bn** semaglutide
- **2 generics launched** (Dr Reddy's, Apotex); **4+ more under review**
- 3rd generic triggers mandatory **65% price cut** → price from **~\$150** → **~\$50/month** by 2027

India (Mass-Market Expansion)



- **13 companies, 26 generics** launched since March 2026
- Prices as low as **~\$15/month** (vials) to **\$60–70/month** (premium injectables)
- Semaglutide volumes up **>6x** (Feb–Apr 2026); generics at **>80% volume share**
- Market scaling from **\$123m (CY25)** → **~\$1bn (CY30)**
- LLY Mounjaro grew **~10%** despite generics (halo effect)

Brazil (Affordability Unlock)



- First generic (EMS/Ozivy) approved **May 2026**; **17 filings** in process
- Current prices **\$160–320/month** → forecast **~\$60/month** by 2027
- GLP-1 market: **~\$5bn** (2028)
- LLY expected to **dominate branded segment** with Mounjaro

China (Delayed but large volume potential)



- Semaglutide patent expired **March 2026**; CSFTA extends exclusivity to **April 2027**
- **148m adult diabetics**; GLP-1 penetration **<10%** for diabetes
- **226 Chinese API suppliers** as of Aug 2024; **74 of 90 FDA DMFs** from China
- LLY investing **\$3bn over 10 years** in China for **Foundayo oral manufacturing**
- Large API manufacturers ramping-up to multi-ton capacity

3 THE GENERIC SEMAGLUTIDE OPPORTUNITY



>100m patients could be supplied globally with only **10–15 tons** of injectable semaglutide API

But oral requires 63–73x more API at equivalent strength: Wegovy Pill requires **~9.1 tons of API per 1m patients** on maintenance vs. **125 kg** for Wegovy injectable



Injectable API supply is manageable — oral semaglutide at scale remains a structural challenge for generics

4 FILL-AND-FINISH COULD BE A BOTTLENECK

Injectable Semaglutide

- **52-374 mg**/patient/year (depending on dose)
- **10-15 tons** of API enough for **>100m** patients globally
- Chinese API industry already at **multi-ton** capacity



Oral Semaglutide

- **Up to 9,125 mg**/patient/-year (maintenance)
- **9.1 tons** of API needed for just **1m** patients on maintenance
- Generic oral production at scale **unlikely near-term**



Fill-and-Finish could be a Bottleneck

- **API production:** manageable for injectables
- **Pen/Device:** multiple providers increasing capacity
- **Fill/Finish Potential bottleneck:** initial F&F providers (OneSource, Gland Pharma) ramping-up capacity; vials as a workaround (**India Gx, US and Brazil** compounders)



5 LLY VS NOVO: COMPETITIVE POSITIONING

ELI LILLY (OW)

- **Mounjaro** (GLP-GIP) differentiated on Ph3 H2H data
- **Foundayo:** true small molecule → globally scalable oral
- Pipeline: **Retatrutide** (GLP-GIP-glucagon), Eloralintide (amylin)
- Incretin franchise: **\$60bn** (2026) → **\$101bn** (2035)
- **~53%** intl share, **~60%** US share — global incretin leader

Lilly

VS.



NOVO NORDISK (EW)

- Semaglutide (GLP-1 only) facing generic erosion
- **Oral Wegovy:** peptide-based → API scaling constraints limit ex-US launch
- Key new launches skewed to **2028-2030+**
- Consensus **\$23bn** sema sales in **2033** likely at risk
- CagriSema (2027), zenagamtide oral (**2030**) — execution dependent

6 THE ORAL GLP-1 INFLECTION

Peptide Oral GLP-1s (e.g., oral semaglutide)

- Require **~63-73x** more API than equivalent injectable doses.
- **Higher efficacy** vs. **Foundayo**
- **Food/drink** restrictions associated with formulation
- Potential supply constraints as global demand scales.



Why It Matters

- Oral therapies may drive the next major phase of **GLP-1 adoption**.
- Manufacturing scalability could become a key competitive differentiator.
- Supply chain design may influence **long-term market share** and **geographic expansion**.



Small-Molecule Oral GLP-1s (e.g., LLY's Foundayo, GRPC's Aleniglipron)

- **True small-molecule therapy** rather than a **peptide**.
- Conventional oral solid-dose manufacturing.
- Lower manufacturing complexity.
- Greater global scaling potential.
- Better positioned for broad international rollout.



Summary of key debates

Why does the first entry of generics constitute a key event in the GLP-1 / diabetes market?

Following the expiration of market exclusivity after the end of the patent protection period, generic versions of semaglutide/Ozempic (Novo Nordisk's GLP-1 for diabetes) have been launched in India and Canada, and will launch in Brazil in the coming weeks; other countries will follow in the coming years with the largest markets (US and Europe) seeing generics entry in the early to mid 2030's. With the cost of manufacturing injectable GLP-1 such as Ozempic and Wegovy relatively low, competition from generics will lead to meaningful price erosion (e.g. >65% price reduction in Canada within the first year). This will have a direct impact on Novo Nordisk's existing GLP-1 franchise (~75% of sales), with the company committed to leverage its manufacturing cost advantage to compete with generic makers on pricing in countries where market exclusivity has expired. Improved affordability of semaglutide should drive greater adoption of GLP-1 across countries, in particular in low-to-medium income countries where these drugs are mostly paid out-of-pocket, which should drive volume expansion for the GLP-1 category. We expect the emergence of a 2-tiered market in countries post-semaglutide patent expiry, with semaglutide becoming an affordable mass-market staple molecule, while oral GLP-1's and next-gen products (GLP-1+) from Lilly, Novo and new entrants (Amgen, Roche, AstraZeneca, Pfizer, Structure Therapeutics amongst others) should compete in a higher value, branded segment.

Why is there a debate on the semaglutide generics supply chain? Semaglutide is a peptide drug, and hence not a traditional small molecule like a statin that is easy to copy and manufacture at scale. The success of Novo Nordisk and Eli Lilly's injectable GLP-1 franchises Ozempic/Wegovy (semaglutide) and Mounjaro/Zepbound (tirzepatide) led to shortages in the US between 2021 and 2024 and limited availability globally. The key bottleneck in the manufacturing process of these drugs were on the Fill-and-Finish (F&F) process, i.e. turning the active ingredient into a sterile liquid and packaging it into a ready-to-use pen or device. Fill-and-Finish requires highly controlled cleanroom space to minimise contamination risk, requires FDA inspection/approval, and building new F&F capacity takes 3+ years. Novo and Lilly invested dozens of US\$ billions in building more capacity for F&F, including Novo's acquisition of three F&F sites from Catalent in 2024. Since then, the combination of manufacturing optimisation and new F&F sites coming online have released the manufacturing bottleneck on injectables, and Novo Nordisk and Lilly have been able to supply the US market and expand the geographic reach of their injectable franchise. However, there is an investor debate on whether generic makers will be able to replicate the supply chain of Novo and Lilly and manufacture enough injectable generics to address the growing demand stimulated by improved affordability. Our analysis suggests that while there is plenty of capacity scaling in China for the API (active pharmaceutical ingredient), the situation is less clear for F&F capacity, and generic makers will likely either need to invest to build capacity in coming years to address the largest markets (US and Europe), or introduce formulations where F&F is less of a bottleneck (e.g. multi-dose vials rather than pen/autoinjectors).

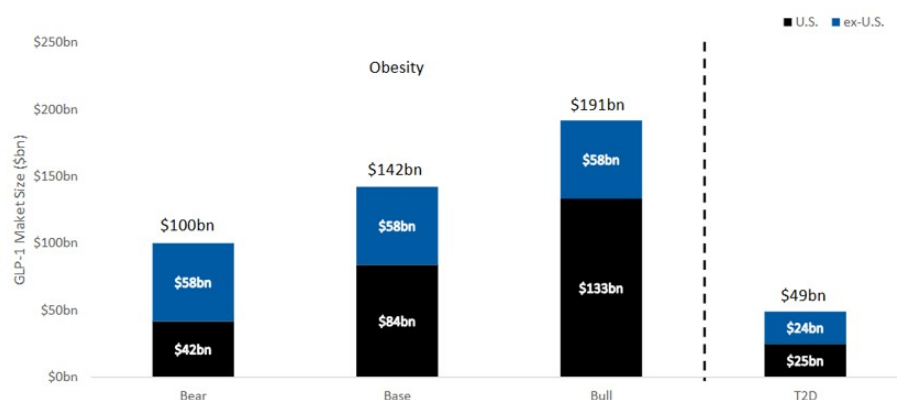
Peptide vs. small molecules, why does it matter in the GLP-1 oral pill battle and in the debate around the generics impact? The answer resides in convenience and scalability. **Novo's oral GLP-1 is a reformulation of semaglutide, a peptide, into a pill form.**

Peptides are a type of molecule which usually cannot be taken orally as they are mostly broken down by stomach enzymes before they can reach their target in the body. To get around this limitation, Novo Nordisk leveraged a formulation technology (SNAC) allowing for improved peptide absorption, but with two drawbacks: 1) the need to take the pill in the morning on an empty stomach and not eat/drink/take other pills for 30 minutes; and 2) the need for a very large quantity of API (active pharmaceutical ingredient) to compensate for a poor absorption through the oral route. Indeed, the pill formulation of semaglutide requires 63-73x more API than the injection for the maintenance dose of equivalent strength of the injectable formulation. This increases manufacturing cost but also limits the scalability of the pill, limiting Novo's reach to the US and some select OUS markets. This also limits the ability of generic manufacturers to scale generic copies of Novo's oral pill. In contrast, **Lilly's oral pill Foundayo (orforglipron) is a true small molecule GLP-1 receptor agonist** which is easier to manufacture at large scale and does not have the fasting/food/drink restriction of the semaglutide oral pill. We note however that cross-trial comparison favours semaglutide in terms of efficacy (weight loss of ~12% for Foundayo at week 72 vs. 16-17% for Wegovy Pill at week 64).

How does the generic debate feed into Morgan Stanley global obesity model forecasts?

In our most recent global report on the GLP-1 broadening (see [Biopharma: Obesity Medications: Tracking the GLP-1 'Unlock' \(7 Apr 2026\)](#)), we framed our expectations for the global GLP-1 market (combining branded and generic drugs) and forecast \$142bn obesity market in a base case (assuming that the branded market retains a higher value through differentiation and market segmentation), and \$100bn in a bear case (including pricing pressure and generic competition). We separately forecast \$49bn market for GLP-1 diabetes in 2035, leading to a global GLP-1 market of \$190bn in our base case (from a ~\$90bn market in 2026). Insights from market dynamics in Canada in particular will inform our view on how other developed markets will evolve in terms of price, penetration/volume, and growth of the oral and innovative branded segments.

Exhibit 2: 2035 GLP-1 Obesity Market: Bull, Base, and Bear Scenarios



Source: Morgan Stanley Research estimates

Key conclusions

1) We believe generic makers will have no issue sourcing API, but fill-and-finish (F&F) could be a bottleneck

Our global deep dive on the emerging supply chain for generic semaglutide (Ozempic) suggests that producing enough API (active pharmaceutical ingredient) will not be an issue for injectable formulations, although scaling supply for the oral pill formulation will remain a challenge. The API requirements for semaglutide injectable is modest, with 10 to 15 tons of API enough to supply >100m patients with Ozempic or Wegovy low dose annually. The Chinese API industry has already set up large chemical synthesis manufacturing capacity for semaglutide API with credible local players already building towards multi-ton capacity. Our analysis of the FDA import data suggests that 40% of semaglutide shipments into the US (ex-Novo/Catalent) over 2024-2026 originated from 5 Chinese suppliers (JYMed, Sinopep-Allsino, Hybio Pharmaceuticals, Fujian Genohope, Aurisco), with a large proportion of shipments likely supplying the US compounding industry. However, supplying a generic oral pill version of semaglutide will be challenging even with multi-ton capacity, with Ozempic/Wegovy pills requiring ~63-73x more API than the equivalent strength in injectable formulation at maintenance. We note that Eli Lilly's Foundayo (oral GLP-1 pill) is a true small molecule and not a peptide, and hence the company will be able to scale globally.

However, fill-and-finish (F&F) capacity could potentially emerge as a bottleneck for injectable generics. While pen/cartridge manufacturers continue to invest and ramp-up capacity and could be integrated into semaglutide generic supply chains, the situation is less clear for F&F capacity, which was also the bottleneck faced by Novo Nordisk and Eli Lilly when ramping-up injectable supply over 2021-2024 before their large investments in F&F (and the Catalent acquisition for Novo) progressively resolved the issue. Investments to address the demand for generic semaglutide have started (e.g. OneSource capacity increase to supply some of the first approved generics), but disclosures from generic companies and CDMOs on F&F capacity allocated to generic semaglutide remain scarce. We believe Canada, India and Brazil are likely to be a testing ground for generic makers to assess market dynamics, and positive ROI in these geographies is likely to trigger larger investments in F&F to address larger markets in Europe (LOE 2031) and the US (LOE from 2032). The launch of vial formulations (three on the market in India and likely to be replicated elsewhere), which are cheaper and easier to manufacture could also be a workaround to the F&F bottleneck, as seen previously with the launch of vials by compounders in the US and Brazil.

2) Generic market formation in Canada, India and Brazil indicate sharp price erosion, volume expansion and emergence of a 2-tier market

The market formation has started in Canada, the second largest GLP-1 market; we expect steep price erosion. Canada will be a key battleground for Novo and generic makers, as it provides the best blueprint for what could happen in the US and Europe in the early/mid-2030s. We estimate that Canada was a \$2-2.5bn GLP-1 market in 2025, including \$1.5-2bn sales for semaglutide. Dr Reddy's and Apotex have launched the first two generics in the market; the launch of a third generic in coming months (likely from

either Sandoz, Teva, Taro or Aspen) would trigger a mandatory 65% price cut on branded price, leading price to decline from ~\$150/month pre-generic to ~\$50/month before the end of 2026. We expect pricing pressure to continue in 2027 as generic makers compete for market share, and we forecast the Canada semaglutide market to stabilise at \$800-900m through a combination of volume growth and price decline, with Novo retaining a ~\$250m tail.

With the entry of generics, the India market for GLP-1 could scale up from \$123m in CY25 to ~\$1bn in CY30. 13 companies have launched 26 generics of semaglutide since March 2026 across several presentations (single-dose, multi-dose, vials and pills) at price points from ~\$15/month for vials to \$60-70/month for more premium injectable offerings. Two players have launched oral tablets (generic Rybelsus) in the 3mg, 7mg and 14mg strength priced ~\$1-\$2.4 per tablet. We expect more launches to enter the market in the coming months. Semaglutide volumes have increased >6x between February and April, with generics already capturing >80% of semaglutide volume, and key players starting to emerge (Torrent, Lupin, DRL, Zydus, and SunPharma). Post generics launch, Eli Lilly continues to be the largest player in value through premium pricing with Mounjaro (GLP-GIP). LLY noted on 1Q26 earnings that the recent launch of generic semaglutide has had a halo effect on the overall GLP-1 market, and Mounjaro has actually grown 10% in the face of generics (see [HERE](#)). Novo Nordisk value and volumes have remained flat in absolute at a low level, with the market expansion driven by generics. We forecast the market to continue to grow from ~\$123m in CY25 to ~\$1bn in CY30, with a segmentation between mass-market volume growth for semaglutide (likely dominated by fewer scaled generics players with supply chain integration, device capabilities, and distribution strength) and a premium market likely dominated by Eli Lilly's Mounjaro.

Brazil's first generic version of semaglutide has been approved; we expect prices to decline followed by market expansion to the benefit of generics, and Eli Lilly in the branded segment. The first generic from local manufacturer EMS was [approved in Brazil on 26 May](#), with a launch expected in June/July and potentially further generic manufacturers to follow (17 generic filings in process). Brazil's branded GLP-1 market is currently underpenetrated, with current prices (\$160-320/month) restricting branded GLP-1 access to higher-income patients in the context of a largely out-of-pocket market; elevated price has also led to the formation of a large compounded/unauthorized import market potentially as large as 3-4x the official market. We forecast the introduction of semaglutide generics to drive down costs (down to ~\$60/month by 2027) and increase penetration, growing the Brazil GLP-1 market from ~\$2bn in 2026 to ~\$8bn in 2028. We expect generic makers (EMS, Hyperia, and global competitors) to be the main beneficiaries at the expense of Novo's franchise, while Eli Lilly should continue to dominate the branded segment with Mounjaro. Risks to our Brazil market forecast could come from further expansion of the compounded/black market.

3) Stock conclusions: prefer Lilly over Novo, prefer generics companies will global reach and integrated supply chains, prefer scaled API suppliers

We expect LLY's (OW rated) franchise to be more insulated from generic pressure given: (1) Differentiation of Mounjaro (Tirzepatide; GLP-GIP) vs. Semaglutide (GLP) based on head-to-head Ph3 data in obesity and T2D, (2) a portfolio of pipeline drugs with novel

mechanisms of action such as Retatrutide (GLP-GIP-glucagon) and Eloralintide (selective amylin), and (3) a small molecule oral GLP-1 pill in Foundayo (notably this is not a peptide like Novo's oral Wegovy, which constrains scalability as we discuss within). LLY's CSO recently noted that Retatrutide could also represent a "workhorse" for obesity patients with lower BMI, as the low dose Ph3 data compare favorably to Tirzepatide with respect to weight loss, tolerability and time of dose escalation (see [HERE](#)). We model LLY incretin franchise sales growing from ~\$60bn in 2026 to ~\$100bn in 2035. See our note from earlier this year [Tracking the GLP-1 'Unlock.'](#)

We expect Novo Nordisk to be under pressure in post-LOE markets. We expect Canada to be the bellwether for future developed market dynamics over coming years; specifically, the extent to which price and volume competition in the semaglutide franchise leads to an erosion of Novo's Wegovy/Ozempic sales. We see some scope for Novo to partially compensate through its Wegovy Pill and through innovation. On Wegovy Pill, script trends in the US YTD have been encouraging (although we are waiting to see prescriptions trends holding through the summer and escalation to highest doses) but we believe the API requirement will limit Novo's ability execute on a broad ex-US launch (unlike LLY with Foundayo). On innovation, CagriSema (GLP-1/amylin) launch in 2027 will be a first step to partially protect the franchise in the US, but we see more potential from the cagrilintide mono (amylin, potential launch in 2028), zenagamtide/amycretin subcut (GLP-1/amylin 2029-2030) and more importantly zenagamtide oral (2030), which carries the highest differentiation potential in our view. At this stage, we see less potential for Novo's triple agonists, which will only come to market next decade and 5+ years after Lilly's retatrutide. Therefore, with key new launches skewed to 2028-2030+, we see a risk for Novo's diabetes/obesity franchise to struggle between inexpensive semaglutide generics and innovation from Lilly and other branded players, and we see consensus' forecasts for \$23bn sales for semaglutide in 2033 as likely at risk (MSe \$16bn).

We highlight potential beneficiaries amongst generic makers / CDMOs / suppliers.

Generics

- **Dr. Reddy's (EW):** early semaglutide generic entrant with integrated API, partner-led pen/fill-finish capacity and aggressive expansion plans, positioning it as a scaled low-cost supplier across Canada, India and other markets.
- **Sandoz (UW):** GLP-1 generics are a modest near-term opportunity via Canada/Brazil launches from 2H26, with upside only if pricing stabilises at an attractive level and if Sandoz is successful in becoming a market share leader. Longer term upside will depend on investments in fill-and-finish to capture the US and Europe opportunity.
- **Hypera (OW):** Brazil semaglutide generics could be meaningful if approval comes in time and launch is strong, with potential market share gains supported by physician access, retail reach and brand-building despite EMS's first-mover/local-manufacturing edge.
- **United Labs (3933.HK, EW):** early China semaglutide generic filer with diabetes sales infrastructure and broader diabetes pipeline optionality, including Novo's ex-China license of UBT251.

APIs/CDMOs/Others

- **Medtide (3880.HK, EW):** has credible semaglutide API capability and FDA import eligibility, but remains more of a peptide/oligo CDMO; although we note that FDA import data suggests that Medtide is already shipping semaglutide API into the US.
- **Raia Drogasil (OW):** lower semaglutide prices should expand Brazil's formal GLP-1 pharmacy market, with RADL well placed to benefit from higher volumes, repeat prescriptions and channel migration from imports/compounding.

Drug Packaging

- **West Pharmaceutical Services (EW):** We expect minimal product trade down from branded to generic GLP-1s, enabling WST to serve the generics GLP-1 market with its high value portfolio, which should drive strong top line flow through to the EPS line. In addition we see the potential for ASP upside from lesser volume discounts as generics players will not be able to receive the same level of volume discounts compared to the two main branded GLP-1 players.
- **Stevanato Group (EW):** We see STVN benefiting from generic GLP-1s for the same aforementioned reasons as WST, but will benefit to a higher degree due to higher company exposure to GLP-1s.

Supply chain formation for generic semaglutide

The emergence of a global semaglutide market will depend on the ability of generic makers to establish and scale supply chains comparable to Novo Nordisk and Eli Lilly.

The shortage of Ozempic/Wegovy/Mounjaro over 2021-2024 has illustrated the complexity in building and scaling-up supply chains to meet global demand for injectable GLP-1. As innovative and generic competition continues to drive a moderation in prices over the coming years, we expect GLP-1 penetration in diabetes and obesity to increase, leading to meaningful volume expansion. Therefore, generic makers seeking to enter and expand global GLP-1 markets with generic semaglutide products will also need to establish and scale supply chains including API, pen/devices and fill-and-finish.

API considerations: likely manageable for injectables, challenging for orals

Semaglutide API requirements depend on formulation, ranging from relatively low API quantity needed for Ozempic 1mg to large volumes required for oral formulations of Ozempic/Wegovy.

- **Ozempic Injectable (52mg/patient/year to 104mg/patient/year):** ramp up dose is 0.25mg/week for 4 weeks in a pen (1mg/multi-dose pen); then 0.5mg for 4 weeks in a pen (2mg multi-dose pen); then maintenance dose either 1mg/week (4mg multi-dose pen) or 2mg/week (8mg multi-dose pen). For patients this implies an API requirement of 52mg/patient/week or 104mg/patient/week depending on the maintenance dose.
- **Wegovy Injectable (125mg/patient/year to 374mg/patient/year for high dose):** Wegovy maintenance dose is 2.4mg/week (standard), to 7.2mg/week (Wegovy High Dose). This implies an API requirement of 125mg/patient/year Wegovy standard or 374mg/patient/year for Wegovy high dose.
- **Wegovy Pill (up to 9.125g/patient/year for a patient on maintenance):** the daily doses for Wegovy Pill are 1.5mg for the starting dose (one month) / 4mg (one month), 9mg (one month) / 25mg (maintenance). At the maintenance dose of 25mg, Wegovy pill requires 9.125g of API/patient/year, 73x more API than Wegovy injectable 2.4mg and 24x more API than Wegovy HD. We note that the proportion of patients ramping to high dose is still a debate currently considering the recent launch of the pill.
- **Rybelsus / Ozempic Pill (up to 9.125g/patient/year for a patient on maintenance):** for semaglutide pill for the diabetes indication, the previous formulation (under the Rybelsus name) was 3mg/7mg/14mg daily; the equivalent under the new brand and the new formulation Ozempic Pill is 1.5mg/4mg/9mg, with Novo working on introducing a higher dose of 25mg. This would simplify the supply chain with Ozempic Pill and Wegovy Pill being effectively the same drug with the same formulation and the same doses (only differing in brand and indication).

Exhibit 3: API quantity required for one patient on Injectable drug for one year (at maximum/maintenance dose)

	Ozempic 1mg	Ozempic 2mg	Wegovy 2.4mg	Wegovy HD
mg/patient/week	1.0	2.0	2.4	7.2
mg/patient/year	52	104	125	374
kg of API for 1m patients on maintenance	52	104	125	374

Source: Morgan Stanley Research

Exhibit 4: Wegovy Pill and Ozempic Pill require much larger amounts of API than the Injectable

Wegovy Pill & Ozempic Pill (new SNAC formulation)		
New patient staying on drug for 1yr		
Daily dose	# days	Total mg
1.5mg	30 days	45
4mg	30 days	120
9mg	30 Days	270
25mg	275 Days	6,875
mg/patient/year		7,310
kg of API for 1m new patients		7,310

Patient staying on maintenance dose for 1 yr		
Daily dose	# days	Total mg
mg/patient/year	365	9,125
kg of API for 1m maintenance patients		9,125

Source: Morgan Stanley Research

API manufacturing: hybrid recombinant / synthetic (Novo) vs. full chemical synthesis (Generics)

Semaglutide can be produced through a semi-recombinant / semi-synthetic approach (Novo) or through chemical synthesis (most generics). Novo's manufacturing route can be described as semi-recombinant / semi-synthetic, with the peptide backbone manufactured through a yeast-based RNA recombinant method, and chemistry introducing additional modifications. In contrast, many generic developers are producing the API through chemical peptide synthesis, usually Solid-Phase Peptide Synthesis (SPPS) or hybrid SPPS and Liquid-Phase Peptide Synthesis (LPPS), followed by side-chain modification and extensive purification. When considering applications from generic makers, regulators focus on whether the final API is the same and whether route-specific impurities are controlled.

The semi-recombinant manufacturing is more expensive to establish (high initial CAPEX requirement) but is more efficient at scale, while the chemical synthesis route has lower barriers to entry but also lower yield at large scale. The key advantages of Novo's approach (yeast-based peptide backbone) is efficiency and better opportunity for yield optimisation at large scale, but building capacity is more capital intensive and takes longer. This is not an easy route for generic makers as it requires biologics/fermentation infrastructure, technology and know-how. By contrast, the chemical synthesis route is faster to develop and the manufacturing chain is more flexible/modulable, but does not have the same yield optimisation opportunity compared to the semi-recombinant route.

We expect API providers and generics makers to be able to supply the API for the injectable formulation at global scale, but producing sufficient API for the pill will be a challenge.

The API required to supply injectable semaglutide is modest; on a rough estimate, the generic industry could supply >100m patients at the global level with only 10-15 tons of API. The API required to supply at large scale the global market with Ozempic 1mg/week (corresponding to the 4mg pen already launched by Dr Reddy's and Apotex in Canada) appears manageable; with 52kg of semaglutide API enough to supply 1m patients at the maintenance dose. The API requirements escalate with the higher doses of semaglutide, with 374kg of API required to supply 1m patients on the Wegovy High Dose pen at maintenance dose. Assuming that the generic industry will focus the generic opportunity on Ozempic and Wegovy, the combined industry could supply >100m patients with 10 to 15 tons of API. We believe the China generic makers, API suppliers and CDMO companies already have significant API capacity and will be able to ramp up to the level required to supply global markets over the coming years.

We believe API and formulation technology considerations are likely to prevent generic manufacturers from manufacturing semaglutide pill generics at scale, although we note that some companies have already launched oral pill (Rybelsus generics) in India. The API requirements for Ozempic Pill and Wegovy Pill are much larger than for the injectable formulations. As an example, to supply 1 million patients on the maintenance dose assuming 100% compliance over a year, 9.125 tons of API are required for Wegovy Pill, 73x more than for Wegovy 2.4mg (125kg for 1m patients on maintenance) and 24x more than for Wegovy High Dose (374kg for 1m patients on maintenance). Additionally, on top of pure API capacity, generic makers would also need to match Novo's updated SNAC technology in terms of API absorption through the oral route. However, we note that generic maker Torrent Pharmaceuticals has launched a generic version of Rybelsus ([Semalix](#)), and Dr Reddy's has launched an oral version ([Obeda](#)), although supply capacity is unknown. Several other pill generic projects are in development for ex-US countries.

Exhibit 5: API quantities

API (Kg) / year required to address the potential market (scenario analysis) (Patient on 1yr maintenance)	# patients on generics (millions)			
	10	50	100	200
API mg/week/patient				
1mg (Ozempic)	520	2,600	5,200	10,400
2mg (Ozempic high dose)	1,040	5,200	10,400	20,800
2.4mg (Wegovy)	1,248	6,240	12,480	24,960
7.2mg (Wegovy HD)	3,744	18,720	37,440	74,880

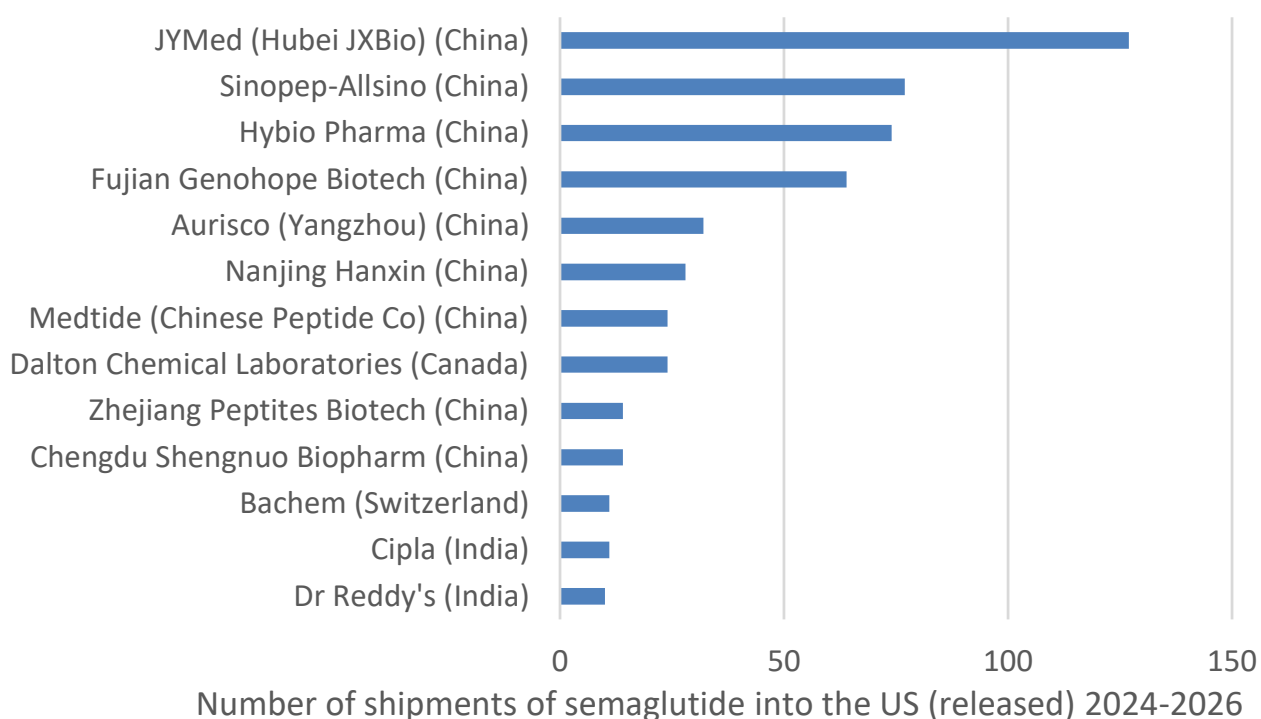
Source: Morgan Stanley Research

China is emerging as a key manufacturing hub for semaglutide API

FDA import data suggest that Chinese API companies are already exporting large quantities of semaglutide API into the US. Analysing the FDA import data over 2024-2026 show that there were a large number of shipments of semaglutide from China and India API suppliers and CDMO into the US. Out of ~2800 released shipments of semaglutide in the US over the period, about a third were from companies other than Novo Nordisk/Catalent. FDA data show that excluding shipments from Novo Nordisk / Catalent, 55% came from China and 14% from India. Albeit a number of these shipments

were for R&D or other legitimate purposes, a large number was also aimed at supplying compounders in the US. The companies with the largest numbers of released shipments according to the FDA data are **1) JYMed (Hubei JXBio), 2) Sinopep-Allsino, 3) Hybio Pharmaceuticals, 4) Fujian Genohope, 5) Aurisco (Yangzhou)** - all manufacturing from China and accounting for ~39% of shipment to the US (excluding Novo/Catalent shipments). We note that **Aurisco** commented in 2025FY earnings that they have a 300 kg/year semaglutide API production line put into production, with further 2026 expansion depending on market demand. **Sinopep-Allsino** also disclosed in 2025FY earnings that they are working on building up to 2.75 tons of semaglutide API capacity / year through their Workshop 601 and 602 projects.

Exhibit 6: Companies importing semaglutide into the US, ranked by number of shipments (ex-Novo)



China has accumulated significant semaglutide API capacity over the years, and is the world's biggest supplier of APIs. China's CDMOs (focusing on novel drugs) and API makers (focusing on generic APIs and formulations) have readied substantial capacity in anticipation of generic launches in China and overseas, although not all producers are choosing to allocate capacity, and commentaries from listed manufacturers suggest a wide range of quality for the semaglutide API.

Partly a result of regulatory loophole: The compound patent for semaglutide patent was filed in March 2006. Under the 20-year patent term set by the WTO, it was set to expire in March 2026, which is what happened in China on March 20, 2026. In the US and EU, however, Novo has pushed out the expiry dates to 2031-2033, leveraging the patent term extension mechanisms there.

- In the US, the Patent Term Adjustments and Patent Term Extensions under the Hatch-Waxman Act offer extensions to compensate for time lost during the regulatory review process. These pushed the expiries of the compound patent (US8129343) and the patent

for use of long-acting GLP-1 peptides for diabetes/obesity (US10335462) to December 2031 and June 2033.

- In Europe, the Supplementary Protection Certificates serve the same purpose, extending patent expiries by up to 5 years. This pushed semaglutide patent protection up to March 2031.

- In China, however, no such mechanism existed prior to June 2021, when the government introduced a similar extension mechanism modeled on US/EU's. Semaglutide (Ozempic) was first approved in China in April, 2021 for the diabetes indication, thus narrowly missing the enactment of the extension pathway. Application of the law is not retroactive either.

Why is China uniquely positioned to manufacture peptides at scale? With the above as background, we distill the following reasons why China has accumulated so much semaglutide capacity, or peptide capacity for that matter.

1. Pre-existing position as the world's API factory: About 30% of the world's API capacity is located in China. In the 20+ years since joining the WTO, China's cost advantages, vast pool of chemistry talent, and a willingness to start from the lower-end of the pharmaceutical value chain have catapulted China to a leadership position in API, with 40+ export-oriented listed API companies ([see our primer](#)). As experts in synthetic chemistry, many have developed solid-phase peptide synthesis capacities.

2. Operating at lower cost: This advantage stems from lower labor costs and raw material prices ([note](#)). This positions China as an attractive service provider to more cost conscious customers.

3. Semaglutide's early patent expiry: A delayed enactment of the patent term extension mechanism led to an expiry date (April 2027) that is 4-5 years ahead of those in US/EU. Coupled with a vast Chinese diabetes population (~12% of the adult population, [source](#)), this created a powerful incentive for Chinese producers to invest in semaglutide API capacity. Some are also eyeing the export markets of Canada, Brazil, and India. As of Q1'26, 74 Chinese companies had obtained FDA DMF registrations for semaglutide API supply, while in China, 13 had filed applications for API supply, with none approved yet. And as showed in [Exhibit 33](#), 10+ companies have submitted applications for biosimilars, but the frontrunners are being held up by the China-Switzerland Free Trade Agreement (CSFTA) ([news](#)).

4. A vast and under-penetrated domestic market: There were 148 million adult diabetics in China as of 2024. Current barriers to uptake in the prescription channel include low levels of reimbursement, needle phobia (see our survey [here](#)), and a gray market which is under increased regulatory crackdowns. In other words, China's API makers could pivot either way: China or overseas.

5. Mastery of SPPS (solid-phase peptide synthesis): SPPS is a widely used lab technique for chemically creating peptides. It strings together amino acids one by one on an insoluble resin bead. Semaglutide is a complex 31-amino acid peptide with a fatty acid side chain, making large-scale, high-purity production technically challenging. Chinese manufacturers have successfully mastered the core technologies for large-scale, high-purity production of long-chain peptides like semaglutide, including technological mastery and process maturity that presents significant technical barriers to entry. A number of

leading manufacturers have achieved ton-scale mass production of semaglutide, with some production lines designed for 10-ton scale.

6. Significant capex in peptide infrastructure: China's leading CDMOs in the TIDES (peptides + oligonucleotides) space have deployed significant capital to capacity buildup. WuXi AppTec is the leader, with SPPS capacity reaching 100,000L by 3Q25 and a projected increase of 30% within 2026 to 130,000L. In 2023 alone, its TIDES business produced more than 15 tons of peptide APIs and intermediates (on 50 preclinical to commercial stage projects). Asymchem had 45,000L of SPPS capacity by end-2025, and expects to expand this to 69,000L within 2026 (+53%). Medtide, a pure-play CDMO specializing in TIDES, has an annual production capacity of over one ton. Other suppliers include Pharmaron, Jiuyuan Genetic, and Sihuan. As of end-2024, Sihuan had two sets of 12,000L fermentation systems dedicated to semaglutide production.

7. Government support: According to commentary ([link](#)) by Caplin Point Labs, supportive policies by the Chinese government have created a highly competitive and fragmented supplier base of 500-600 companies in the biosimilar/generic space, versus 7-10 large companies in India.

8. Sizable export markets: What originally incentivized the domestic supplier base was China's own generic market (now delayed to April 2027) and the compounding market in the US. Following restrictions in the US on the latter, Chinese suppliers have since pivoted towards other generic markets in addition to China's, e.g. Canada, Brazil, and India.

Quantifying the domestic base for semaglutide API manufacturing: CDMOs and API makers alike have ramped up significantly their peptide capacities. As of Q1'26, the US FDA had granted 90 DMFs (Drug Master Files) to suppliers for the semaglutide API. Of those, 74 were Chinese companies (82%), followed by 8 for India (9%). Despite a large number of Chinese suppliers for semaglutide, only a few are on FDA's Green List. Medtide (3880-HK) has disclosed that it is one qualified supplier - *"The Group's semaglutide API was included on the Green List of U.S. FDA Import Alert 66-80, designating it as eligible for entry into the United States without detention. This recognition follows FDA's evaluation of the Group's manufacturing facilities and quality systems, and reflects the Group's compliance with U.S. cGMP requirements."* The Green List comprises foreign manufacturers whose facilities and products have been evaluated by the FDA and comply with current Good Manufacturing Practice (cGMP) requirements. In terms of peptide capacity, the largest suppliers in China are CDMOs WuXi AppTec and Asymchem, although these companies service mainly novel drug contracts. A number of suppliers have ton-scale annual capacities for peptide APIs, but not all have significant plans for semaglutide. We set out the potential semaglutide suppliers below. JYMed, Hybio, and Jiangsu Sinopep-Allsino are openly committed to semaglutide API supplies. Medtide has ton-scale peptide capacity but has not commented on an intent to capture the semaglutide market (although FDA shipment data suggest that the company is exporting batches to the US through its subsidiary China Peptide Company). For now, it is focused on novel drug development.

Exhibit 7: Peptide and known semaglutide suppliers (highlighted rows)

Company	Ticker	Role	Key Capacity Metric	Quantity Disclosed
WuXi AppTec	603259.SS / 02359.HK	CDMO (tirzepatide primary supplier)	SPPS Reactor Volume / Annual Production	>100,000L SPPS (end-2025); >130,000L projected 2026; ~58+ metric tons peptide APIs delivered in 2025
Asymchem	002821.SZ / 6821.HK	CDMO (semaglutide DMF filed)	SPPS Reactor Volume / Annual Capacity	45,000L SPPS (end-2025); >22.5 metric tons/year; expanding to 69,000L by end-2026
Medtide	3880.HK	CDMO (semaglutide on FDA Green List)	Annual API Capacity / Batch Size	>1 metric ton/year; >50 kg/batch (Hangzhou); 300 kg/year planned (U.S.)
JYMed Technology	Private	API supplier (semaglutide DMF filed)	Annual Capacity	>4 metric tons/year (Hubei); expanding to >10 metric tons by 2026
Hybio Pharmaceutical	300199.SZ	API supplier (NMPA-registered semaglutide)	Batch Size / Annual Capacity	>10 kg validated batches (semaglutide); 5,000–10,000 kg/year projected (all peptides post-renovation)
Sihuan / Huisheng Biopharm	460.HK	Biosimilar developer	Fermentation Systems / Injection Capacity	Two sets of 12,000-liter fermentation systems; >100 million injection units (Phase II design)
Sichuan Hexie Shuangma / Jianyuan	000935.SZ	API supplier / CDMO	N/A (no specific figures disclosed)	40+ peptide API products; 10 U.S. FDA DMFs including semaglutide
Pharmaron	03759.HK / 300759.SZ	CDMO (building capacity)	N/A (under construction)	New larger-scale SPPS facility expected to complete in 2026; no specific figures disclosed
Hangzhou Jiuyuan	2566.HK	Biosimilar developer (Jiyoutai® / Jikeqin®)	N/A (capacity building underway)	Production line planning finalized; no specific figures disclosed
Livzon Pharmaceutical	000513.SZ	Biosimilar developer	N/A (project under construction)	Semaglutide Project listed as construction-in-progress; no specific capacity figures
CSPC Pharmaceutical	1093.HK	Biosimilar developer	Overall: ~30bn tablets/year, ~3bn injectable doses/year	No semaglutide-specific API capacity figures disclosed
Jiangsu Sinopec-Allsino	688076-SH	API supplier	N/A	Major semaglutide API supplier; no specific capacity figures disclosed

Note: Orange denotes open intent to capture semaglutide API market.
Source: Company data; Morgan Stanley Research

Fill-and-finish could be an initial bottleneck for the generics industry

The device requirement based on the multi-dose pen is 13 pens/patient/year. For Ozempic, Novo Nordisk is using the flextouch pen (4 weekly doses in a pen) globally. For Wegovy, Novo Nordisk is using the flextouch pen ex-US and a single-use weekly pen in the US (52 pens/patient/year), although the company is waiting for the approval of the Wegovy multi-dose pen in the US in Q2'26. Therefore, the pen volume requirement for generic maker will likely remain at 13 pen/patient/year across geographies with a multi-dose pen.

Vial formulation can offer lower costs and higher flexibility and could help with supply chain bottlenecks. Ozempic and Wegovy are currently only available in pre-filled pen formulations; however Novo is working on introducing a vial formulation of Wegovy in 2026. The vial formulation (vial + a separate needle and syringe) is an important consideration as it allows for lower manufacturing costs, and more importantly is a way to overcome potential supply chain bottleneck on the pen/device and associated F&F capacity. We note that three generics makers have already launched vial formulations in India, with additional companies expressing an interest in the formulation. In Canada, at least one company (Vimy Pharma) is also planning to sell a vial formulation. We note that grey market / compounded GLP-1 in the US and Brazil are mostly marketed through a vial formulation as well.

Novo's total F&F/device capacity is not known, however we estimate that Ozempic and Wegovy global sales in 2025 corresponded to an equivalent of ~128m multi-dose pens annually; we forecast ~138m equivalent multi-dose pen to support 2026 sales. For Ozempic, we estimate that Novo sold ~100m pens in 2025; we used international sales

(\$5.8bn in 2025) and assumed an average price/pen of \$85 leading to 69m pens for International operations. For the US, we used the TRx number (30m in 2025). We note that in March 2025, Novo indicated that 7m patients worldwide were being treated with Ozempic, which would correspond to 91m annualised pen sales in March 2025, consistent with our estimate of ~100m for 2025. For **Wegovy**, we do a similar calculation assuming \$250/month in IO leading to 17m multi-dose pens. In the US, Wegovy is sold as single-dose pen (4 pen/month vs. 1 pen/month for the multi-dose), so 13m scripts in 2025 correspond to ~51m pens, leading to a total number of pens at 68m. Converting the Wegovy US into equivalent multi-dose pen volume would lead to total multi-dose pen volume of 30m pens for Wegovy in 2025. **Therefore, we estimate that to address Novo's current semaglutide global market, ~128m multi-dose pens would be required.**

Exhibit 8: MS estimate - volume of pens to supply previous semaglutide demand

Ozempic	2023	2024	2025	2026e
Ozempic IO sales (DKKm)	32,708	36,140	38,622	37,035
Ozempic IO sales (\$m)	4,746	5,241	5,837	5,799
Estimated \$/pen	90	90	85	80
Estimated # pen (m) for IO	53	58	69	72
Ozempic US TRx (m)	22	29	30	26
Estimated # pen (m) for US	22	29	30	26
Ozempic global # multi-dose pen (m)	75	87	99	98

Wegovy	2023	2024	2025	2026e
Wegovy IO sales (DKKm)	1,913	12,436	28,091	37,475
Wegovy IO sales (\$m)	278	1,804	4,246	5,868
Estimated \$/pen	350	275	250	240
estimated # multi-dose pen (m) for IO	1	7	17	24
Wegovy US TRx (m)	5	9	13	15
# pen / TRx	4	4	4	4
Estimated # single-use pen (m) for US	19	36	51	60

Wegovy + Ozempic	2023	2024	2025	2026
Estimated # multi-dose pen (m)	76	93	116	123
Estimated # single-use pen (m)	19	36	51	60
MS estimated # pen (m) total	95	130	167	183

Assuming US Wegovy on multidose:

MS estimated # multi-dose pen (m)	80	102	128	138
--	-----------	------------	------------	------------

Source: Morgan Stanley Research

Novo Nordisk has made large investments in fill-and-finish capacity in recent years, including the acquisition of 3 fill-and-finish sites from Catalent. Novo invested \$31bn in capacity expansion over 2021-2026 (including the \$11.7bn acquisition of 3 Catalent fill-and-finish sites). Of these, around \$18bn was for injectable fill-and-finish (with the majority being the Catalent side acquisition), adding to the existing capacity the company already had for insulin and GLP-1. Novo commented that following these investments, they have enough capacity to comfortably supply the main global markets with injectables, with incremental investment in GLP-1 supply likely to focus on pill supply (e.g. the \$500m investment in Ireland in March 2026). Novo estimate that on average, the construction of an API site takes 5+ years, and fill-and-finish takes 3+ years; therefore the supply from

large investments announced over 2023-2024 should come online over 2026-2029.

Exhibit 9: Novo Nordisk API/F&F investments since December 2021 - \$18bn in F&F including Catalent

Date	Site	Scope	Investment (DKKbn)	Investment (\$bn)
Dec-21	Kalundborg, Denmark	Mainly API	17	2.6
Nov-22	Bagsværd, Denmark	Clinical API	5	0.7
Jun-23	Hillerød, Denmark	Mainly API	16	2.3
Nov-23	Kalundborg, Denmark	Mainly API	42	6.1
Nov-23	Chartres, France	Fill-Finish	16	2.3
Dec-23	Athlone, Ireland	Oral Portfolio	1	0.1
Jun-24	Clayton, US	Fill-Finish	27	3.9
Dec-24	Odense, Denmark	Finished production (RD)	9	1.3
Dec-24	Catalent (3 sites)	Fill-Finish	82	11.7
Mar-26	Athlone, Ireland	Oral Portfolio	3	0.5
Total			218	31
Of which Fill & Finish			125	18

Source: Novo Nordisk

Commentary from generics and CDMOs on pen / fill-and-finish capacity are scarce and existing capacity in the industry is difficult to assess, but we believe that more investment will be needed to supply globally.

The most vocal F&F company working with generic manufacturers is **OneSource Specialty Pharma**, which is providing fill-and-finish services for Dr Reddy's (Canada) and Apotex (Canada and US). The company is currently executing on a \$100m investment plan to scale its total drug-device cartridge capacity from 40 million to 220 million units/year, with the expansion driven by anticipated demand for semaglutide generic F&F across global markets (only part of the capacity is allocated to semaglutide). **Gland Pharma** is another example of a company investing to build F&F capacity to serve the generic industry, with the company planning to commercialize around 20 million pens and cartridges for GLP-1 products by fiscal year 2027, and has invested to increase its total cartridge fill-and-finish capacity to 140 million units in 2026. We note that some of the largest F&F suppliers (e.g. Thermo Fisher, Simtra and Vetter) are unlikely to work with semaglutide generics makers as they tend to reserve their capacity for higher margins innovator/complex projects.

Commentary from companies involved in pen/cartridge/device manufacturing suggest growing investments and a ramp-up in pen capacity.

Looking at comments from pen/device manufacturers, **Shaily Engineering Plastics** (supplying the pen for Dr Reddy's Canada generic and Apotex US and Canada generics) recently integrated a 25 million pen capacity line dedicated to semaglutide. The company also announced an investment in a new facility in Abu Dhabi to produce 75m pens; and expect to reach a total pen capacity of 150m annually by March 2028 (although not all the capacity will be dedicated to semaglutide generics). The main pen/device manufacturers such as SHL medical, Ypsomed, Medmix, Nemera, Gerresheimer, BD, Schott Pharma, Phillips Medisize, etc. also offer pen/cartridges for GLP-1 supply chains and are likely to be integrated into generics F&F supply chains. Most of these companies have announced capacity expansion plans (often > \$100m) at least partially driven by demand for GLP-1 therapies.

As an example, **Stevanato** (covered by Kallum Titchmarsh) provides drug delivery systems for GLP-1. GLP-1 revenue accounted for approximately 19%–20% of STVN's total company revenue, growing more than 50% compared with 2024. We estimate approximately 90%

STVN's GLP-1 business is branded, with roughly 10% coming from players other than the two largest branded manufacturers (Novo Nordisk and Eli Lilly). The Nexa syringe was STVN's fastest growing product in 2025, driven primarily by GLP-1 demand. The company's strategy is to be present across both originators and biosimilars, noting that historically when a product goes off-patent, approximately 70% of revenue stays with the originator and 30% moves to biosimilars.

Canada generic market formation (2026)

We estimate the existing GLP-1 market in Canada at ~US\$2-2.5bn, of which ~US\$1.5-2bn for semaglutide. Novo Nordisk reported ~US\$1.15bn sales for semaglutide in Canada in 2024 (US\$1bn for Ozempic and US\$150m for Wegovy). While Novo did not disclose semaglutide Canada sales in 2025, we estimate that sales were ~US\$1.6bn. This is consistent with the Dr Reddy's management estimate of Canada's existing semaglutide market of ~US\$1.8bn based on IQVIA data. Including Mounjaro sales (undisclosed, launched in November 2023, full supply availability in May 2025), we estimate the Canada pre-LOE market is ~US\$2-2.5bn.

We see Canada as a blueprint for how market dynamics could evolve in the US and Europe after LOE. Novo Nordisk management indicated that India is not an appropriate blueprint for other countries. One key reason is the under-development of the branded GLP-1 market ahead of generic launch, with only ~US\$50m of combined sales in India for Wegovy/Ozempic due to a combination of supply limitation (Wegovy launched in June 2025 and Ozempic only launched in December 2025, although Rybelsus was launched in 2022) and affordability barriers. Therefore, generic makers have entered India before the branded market could be established, forcing Novo Nordisk to cut prices to remain competitive in the context of generic launches. The second key point is the average income differential between India and the Western market making affordability and price a more acute focus in the country. The third key point is the more intense price competition from a large number of local players approved in India (13+ companies approved at IP expiry in the country). In comparison, Canada is much more comparable to the US and Europe in terms of regulatory requirements, average income and maturity of the branded GLP-1 market (Canada is the second largest GLP-1 market in the world).

Dr Reddy's and Apotex products were the first two semaglutide generics approved in Canada. Both companies received approval for a multi-dose 2mg and 4mg pen (corresponding to the 1.0mg/weekly Ozempic dose) and are approved only for the diabetes indication. The API for both products is manufactured through solid-phase synthetic chemistry (i.e. purely chemical process vs. Novo's hybrid biologic/chemical process). **Dr Reddy's** (approved 28 April, launched 16 May 2026) [generic](#) API is produced in-house. **Apotex** (approved 1 May, launched 14 May 2026) [generic](#) was developed in partnership with Orbicular Pharmaceutical Technologies; while the source of the API is unknown, we believe it is likely supplied by the Orbicular CMO network. The fill-and-finish for both Apotex and Dr Reddy's is handled by OneSource with device components provided by Shaily Engineering Plastics. The Apotex semaglutide generic (developed and manufactured with the same partners) also cleared tentative approval in the US.

Hims & Hers (covered by Craig Hettenbach) has partnered with Apotex to replicate the successful consumer-led approach taken in the US in Canada. H&H has [announced](#) the marketing of Apotex' generic through its platform for weight loss, despite the generic label indication being restricted to T2D. H&H offers treatment options starting at 149 CAD/month (US\$107/month) and follows the company's acquisition of Livewell, which was announced in December 2025. Livewell was founded in 2023 as a telehealth platform for weight loss, smoking cessation and sexual health and this acquisition was specifically aimed to align with the timeline of generic semaglutide availability in Canada. H&H press

release indicates: *Generic semaglutide is authorized for sale by Health Canada for the management of Type 2 diabetes. It may be prescribed off-label for weight management in a licensed healthcare provider's sole clinical judgment. Hims & Hers Canada Inc. does not currently offer access to treatment services for Type 2 diabetes.* We remain Equal-weight on HIMS on a balanced risk/reward, with pressure on margins and decelerating core growth vs. potential upside in international growth and new category ramps (see [Higher revenue guidance comes with a cost, as EBITDA is cut on weaker margins](#)).

We expect the price of Ozempic in the public reimbursement channel to decline from ~US\$150/pen (depending on the region) to US\$70-80 in the near term, then ~US\$50 after a 3rd generic entry. The launch of a 3rd generic in Canada will trigger a price cut for all Ozempic generics in the reimbursed channels. Generic public reimbursement/list prices in Canada are set through the pan-Canadian Pharmaceutical Alliance (pCPA) Tiered Pricing Framework for generics. When two generics are launched, the generic price is set to 50% of the brand reference price; when a third generic is launched, this triggers an automatic reimbursement/list-price tier change and the generic price is set at 35% of the brand reference price (for an injectable). While Novo Nordisk is not required to lower the price of branded Ozempic to the level of generics, a substantial price cut is likely for the drug to remain competitive against generics. We also note that while private plans in Canada do not have to follow the pCPA price, many do, effectively leading to a spillover of the public price into the private market.

Health Canada is currently reviewing generics from four other generic makers, with additional companies expected to file in coming months. In addition to the two approved generics from Apotex and Dr Reddy's, Health Canada is currently reviewing seven additional generics, of which three are additional filings from Apotex, and the other four are generics from Sandoz, Teva, Taro and Aspen. Aspen indicated that they expect approval between May and September 2026; Sandoz expect approval in the second half of 2026; Teva and Taro have not commented. Three additional companies are also developing a generic of semaglutide in Canada but have not yet filed according to Health Canada disclosures: PharmaTher (targeting approval in 2026), Biocon, and Canadian company Vimy Pharma, which is seeking to bring to market a vial generic manufactured by partners in Canada.

Exhibit 10: Emerging landscape for semaglutide generics in Canada

Company	Status	Information on API / F&F
Dr Reddy's	Generic launched 16 May	API in-house, F&F from OneSource
Apotex	Generic launched 14 May; 3 other semaglutide generics under review	Developped with Orbicular, API likely from Orbicular CMO network; F&F from OneSource
Sandoz	Generic under review; launch planned for H2'26	Partnered for API and F&F; partnered with Pharmathen
Taro	Generic under review	
Aspen	Generic under review; approval expected between May and September 2026	API and F&F in-house
Teva	Generic under review	Unconfirmed, likely in-house
PharmaTher	Generic not yet filed; approval expected in 2026	F&F by an undisclosed partner
Vimy Pharma	Generic not yet filed; filing targeted for 2026; targeting a vial formulation	Local manufacturing in Canada partnered with Dalton and Applied Pharmaceutical Innovation
Biocon	Biocon management indicates generic has been filed and aim for progress in 202; no mention from Health Canada	

Source: Morgan Stanley Research, Company disclosures, Health Canada

Modelling Canada generic impact: We model Ozempic price to decline from US\$150/month pre-generic entry to ~US\$50/month by 2027, before price decline to US\$25/month over the following years. We assume that GLP-1 penetration in the diabetic and overweight/obese population increases from ~7% in 2025 to ~25% in 2035, including 16.5% for Ozempic and generics. We assume that generic makers capture 80% of Ozempic volumes in the long term. This leads to Novo Nordisk Ozempic/Wegovy sales declining from US\$1.6bn in 2025 to ~US\$280m in 2035. Our modelling suggests a -1% impact on group sales from generics entry in Canada in 2026, consistent with Novo's comment for -LSD% impact on group sales from LOE in all countries (of which Canada is the largest).

Exhibit 11: Modelling of the Ozempic/Ozempic generic market in Canada

	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
Adult population Canada (m)	33.7	34	34.3	34.7	35.0	35.4	35.7	36.1	36.5	36.8	37.2	37.6
% Overweight/Obese/Diabetic	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%	66.0%
Eligible adult Canada patients	22.2	22.4	22.7	22.9	23.1	23.4	23.6	23.8	24.1	24.3	24.5	24.8
% GLP-1 penetration	5.1%	7.2%	10.0%	15.0%	18.0%	20.0%	21.0%	22.0%	23.0%	24.0%	25.0%	25.0%
% other GLP1 (tirze, orfo, other)		0.5%	1.5%	3.5%	5.5%	7.5%	8.0%	8.0%	8.0%	8.0%	8.0%	8.0%
% on Wegovy	0.4%	0.7%	0.7%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%
% on Ozempic & Generics	4.7%	5.9%	7.8%	11.0%	12.0%	12.0%	12.5%	13.5%	14.5%	15.5%	16.5%	16.5%
Novo Ozempic Volume share	100%	100%	70%	50%	40%	35%	30%	25%	20%	20%	20%	20%
Generic Ozempic Volume share	0%	0%	30%	50%	60%	65%	70%	75%	80%	80%	80%	80%
Patients on Ozempic/Gx in the year	1.1	1.3	1.8	2.5	2.8	2.8	2.9	3.2	3.5	3.8	4.0	4.1
Stay time (months)	6.0	6.0	6.5	7.0	7.5	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Equivalent number of full-year patients (m)	0.5	0.7	1.0	1.5	1.7	1.9	2.0	2.1	2.3	2.5	2.7	2.7
Average sema pen price (/month) - USD	150	150	93	47	38	32	29	28	27	26	26	25
% change		0%	-38%	-49%	-20%	-15%	-10%	-3%	-3%	-3%	-3%	-3%
Ozempic + Gx Volumes (million pens)	6.9	8.7	12.4	19.1	22.5	24.3	25.6	27.9	30.2	32.6	35.1	35.4
% chg		26.5%	43.0%	54.1%	18.1%	7.7%	5.2%	9.1%	8.5%	8.0%	7.5%	1.0%
Canada sales (\$m)												
Novo - Ozempic (USDm)	1028	1300	808	451	341	273	222	195	165	172	180	176
% change		26.5%	(37.9%)	(44.2%)	(24.4%)	(19.9%)	(18.8%)	(11.8%)	(15.8%)	4.7%	4.3%	(2.0%)
Wegovy sales (USDm)	147	300	328	130	94	101	102	103	104	105	106	107
		104.6%	9.4%	(60.3%)	(27.9%)	7.7%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Novo inj semaglutide sales (USDm)	1174	1600	1136	581	435	374	324	299	269	278	286	283
Generic makers (USDm)			195	451	511	507	517	586	658	689	719	704
				131.1%	13.3%	(0.8%)	2.0%	13.4%	12.2%	4.7%	4.3%	(2.0%)
Total semaglutide market Canada	1174	1600	1331	1032	946	881	841	885	927	967	1005	988
Wegovy assumptions												
Wegovy average pen price (MSe)	280	280	280	150	100	100	100	100	100	100	100	100
Wegovy Volume (million pens)	0.52	1.07	1.17	0.87	0.94	1.01	1.02	1.03	1.04	1.05	1.06	1.07
# full-year patients (m)	0.04	0.08	0.09	0.07	0.07	0.08	0.08	0.08	0.08	0.08	0.08	0.08
Stay time (months)	6	6	6.5	7.0	7.5	8.0	8.0	8.0	8.0	8.0	8.0	8.0
Patients on drug in the year	0.08	0.16	0.17	0.11	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12

Source: Morgan Stanley Research estimates

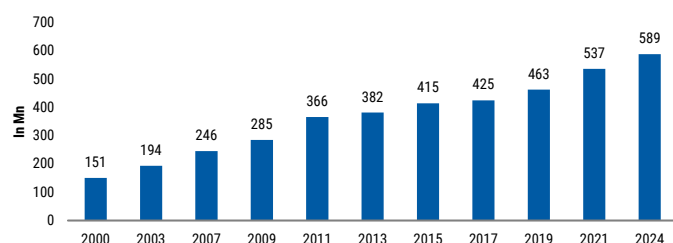
India generic market formation (2026)

- ICMR-INDIAB (2023): India's diabetes prevalence is 11.4%; 40% of adults have abdominal obesity, and 136mn are pre-diabetic. As per National Family Health Survey-5, almost one in four Indians between the age group 15-49 is now overweight or obese.
- Post-LOE in March 2026, semaglutide volumes surged (~6x), with generics capturing large part of injectable semaglutide share; stable innovator volumes suggest growth is driven by new patient addition, not just switching.
- We expect India's GLP-1 market to reach Rs 96.6bn (US\$1bn) in 2030.
- **As of April 2026, Torrent, DRL, Zydus, and Sun have emerged as key players in generic semaglutide.**
- Some Indian players are making API in-house and are actively planning backward integration. Dr Reddy's, in addition to the India market, also has approval for Canada and targets an output of >40mn pens annually by FY28. Divi's aims to be one of the largest players in peptides globally, and currently has the largest SPPS scale (Solid-Phase Peptide Synthesis ie multiple ~3,000L reactors) scale in India and we believe the company is part of Lilly's supply chain. CDMOs such as One Source Specialty and Shaily Engineering plan 4-5x increase in output, whereas Anthem Biosciences plans to manufacture semaglutide API.

We view the semaglutide LOE as a inflection point that transitions India's GLP-1 market from a premium, access-constrained category to a mass-market, volume-driven segment. Early data post-March 2026 shows a sharp acceleration in volumes driven by generics, with affordability unlocking demand beyond affluent urban cohorts and expanding adoption across Tier I/II cities.

Over the medium term, we see : 1) **Expansion Phase (next 12 months)** market expansion continues and more players enter the market in coming months. 2) **Consolidation phase** among fewer scaled players, particularly those with backward integration, device capabilities, and distribution strength. 3) **Segmentation:** Clear segmentation of the GLP-1 market, with semaglutide anchoring mass-market volume growth and Eli Lilly's Mounjaro/tirzepatide retaining a premium positioning.

Overall, we expect the India GLP-1 market to scale sharply from US\$166mn in FY26 to US \$1bn by FY30 [Exhibit 13](#) driven primarily by generic-led penetration gains. The key investor debate shifts from "how big the market can get" to "who captures value in a volume-led, price-competitive environment," with execution on cost, supply chain integration, and brand/distribution emerging as the critical differentiators.

Exhibit 12: Global prevalence of diabetes in the 20–79 year age group (mn)

Source: IDF Diabetes Atlas 11th Edition - 2025, Morgan Stanley Research

Exhibit 13: Prevalence of diabetes country wise (20–79 years) with diabetes in 2024**Top 5 Countries****Top five countries for number of people with diabetes (20–79 years) (In Mn)**

Country	2024	2050
India	90	157
Bangladesh	14	23
Sri Lanka	1.6	1.9
Nepal	1.3	2.4
Mauritius	0.2	0.2

Top five countries for age-standardised prevalence of people with diabetes (20–79 years) (In %)

Country	2024	2050
Mauritius	20.1%	23.5%
Bangladesh	13.2%	15.4%
Bhutan	12.0%	13.8%
India	10.5%	12.8%
Sri Lanka	10.2%	12.0%

Source: IDF Diabetes Atlas 11th Edition - 2025, Morgan Stanley Research

India GLP-1 market

Pre LOE: Eli Lilly's Mounjaro was launched in March 2025 and rapidly scaled up, reflecting strong demand despite premium pricing. By end of FY26 (31 March 2026), it became the largest selling branded drug in India with annual sales of Rs 9.8bn (US\$111mn) in FY26. Novo's Wegovy/Ozempic (semaglutide) initially saw slower uptake due to pricing and supply constraints but gained traction following price rationalisation and distribution partnerships. The overall GLP1 market size in FY26 was US\$166mn.

Post launch of generic semaglutide, the market has expanded and April sales annualised imply market size of US\$275mn, within that generic share was at 20%. With generic semaglutide, we expect affordability to improve materially, expanding adoption beyond affluent urban segments into Tier-I/II cities.

April was the first month post LOE, early indications point:

- Tirzepatide (US\$104mn in FY26) remains the dominant molecule. Cipla (in partnership with Lilly) has scaled to ~10–13% share at peak and Lilly's market share was 63%
- Novo Nordisk remains the second-largest player with ~19–25% share, supported by its semaglutide franchise (Ozempic/Wegovy/Rybelsus) - see [Exhibit 19](#) . [Exhibit 19](#)
- Growth is volume-led with generics driving a 2–3x jump in units and injectables surging ~6x, while value growth (<2x) lags due to price erosion, highlighting an access-led expansion.
- Injectables are outperforming orals - see [Exhibit 24](#) .

Exhibit 14: GLP-1 market model in India

US\$m	FY25	FY26	FY27e	FY28e	FY29e	FY30e
India - GLP - 1						
India population estimated in 2025 (in mn)	1,440	1,441.4	1,442.9	1,444.3	1,445.8	1,447.2
<i>Growth</i>		0.1%	0.1%	0.1%	0.1%	0.1%
Adult population (age 18+) (in mn)	1,008	1,009	1,010	1,011	1,012	1,013
<i>% of population</i>	70.0%	70.0%	70.0%	70.0%	70.0%	70.0%
Overall obesity prevalence among adults (in mn)	81	81	82	83	84	85
<i>% of adult population</i>		1%	1%	1%	1%	1%
Population who could afford treatment (In Mn)		0.1	0.5	1.0	1.3	1.6
<i>% of obese population</i>		0.1%	0.6%	1.2%	1.6%	1.9%
Annual cost of therapy (Rs)		180,000	70,000	66,500	63,175	60,016
<i>Annual price increase %</i>			3%	-5%	-5%	-5%
Market size GLP-1 drug (in Rs mn)		15,780	34,550	66,301	84,821	96,645
In US \$ mn		166	364	698	893	1,017

Source: Pharamtrack, Morgan Stanley Research estimates (e), USD/INR Rate of 95 for all periods

Outlook: Near term, the GLP-1 market is in a hyper-growth phase with strong April 2026 momentum and continued aggressive launches, while over the medium term growth should moderate with emerging consolidation as competition intensifies; regulatory actions may curb misuse but are unlikely to disrupt the structural growth trajectory.

Overall, we expect the India GLP-1 market to scale sharply from US\$166mn in FY26 to ~US\$1bn by FY30, driven primarily by rapid uptake of generics.

Exhibit 15: India GLP-1 market in US\$m

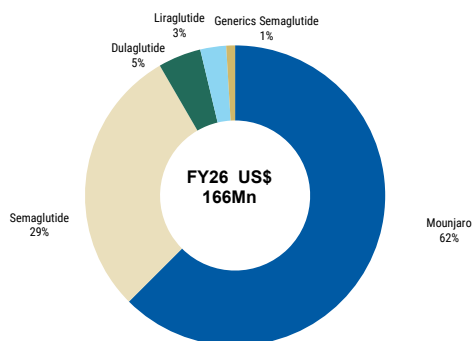
GLP-1 (US\$ Mn)	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Mounjaro	0.0	0.0	0.8	1.6	2.4	3.9	4.5	8.7	8.4	10.5	11.4	12.4	13.8	14.2	12.0	12.7
Semaglutide	3.9	3.9	3.1	3.6	3.4	3.3	4.0	4.2	4.1	3.7	4.2	4.1	4.2	4.9	4.7	4.6
Dulaglutide	0.7	0.7	0.6	0.7	0.7	0.6	0.7	0.7	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.6
Liraglutide	0.3	0.3	0.4	0.4	0.4	0.4	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.3	0.3
Generics Semaglutide	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	1.4	4.6
GLP-1 Agonist Market	4.9	4.9	4.9	6.3	6.9	8.2	9.8	14.1	13.6	15.3	16.5	17.5	18.9	20.1	18.9	22.9

Source: Pharamtrack, Morgan Stanley Research, USD/INR rate of 95 for entire period

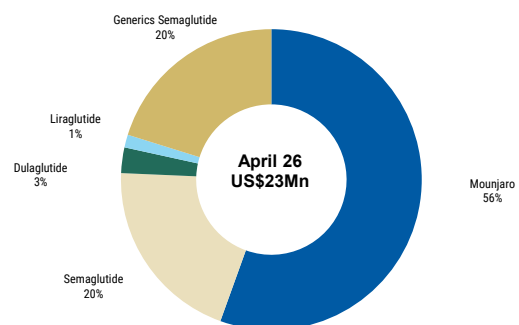
Exhibit 16: India GLP-1 market in Rs mn

GLP-1 (Rs Mn)	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Mounjaro	0	0	80	150	230	370	430	830	800	1000	1080	1180	1310	1350	1140	1210
Semaglutide	370	370	290	340	320	310	380	400	390	350	400	390	400	470	450	440
Dulaglutide	70	70	60	70	70	60	70	60	60	60	60	60	50	50	50	60
Liraglutide	30	30	40	40	40	40	50	40	40	40	30	30	30	30	30	30
Generics Semaglutide	0	0	0	0	0	0	0	0	0	0	0	0	10	10	130	440
GLP-1 Agonist Market	470	470	470	600	660	780	930	1340	1290	1450	1570	1660	1800	1910	1800	2180

Source: Pharamtrack, Morgan Stanley Research

Exhibit 17: Share of GLP-1 market in India (FY26)

Source: Pharmarack, Morgan Stanley Research

Exhibit 18: Share of GLP-1 market in India (April 26)

Source: Pharmarack, Morgan Stanley Research

Exhibit 19:

GLP-1 market share of top 10 Players

Top 10 players in GLP-1 market	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Eli Lilly	69%	69%	63%	63%	61%	56%	47%
Novo Nordisk	24%	25%	23%	22%	25%	25%	19%
Cipla (Yurpeak + Trulicity)	3%	3%	11%	13%	12%	10%	11%
Torrent	0%	0%	0%	0%	0%	3%	8%
Lupin	1%	1%	1%	1%	0%	1%	2%
Eris Ls	2%	1%	1%	1%	1%	1%	2%
Dr. Reddys Labs	0%	0%	0%	0%	0%	1%	2%
Zydus Healthcare	0%	0%	0%	0%	0%	1%	2%
Glenmark	1%	1%	1%	1%	1%	1%	1%
Sun Pharma	0%	0%	0%	0%	0%	1%	1%
Others	0%	0%	0%	0%	0%	0%	5%

Source: Pharmarack, Morgan Stanley Research

Exhibit 20: Generics drug pricing by players

Company	In-house/ Partner	Brand(s)	Indication	Dosage Strengths*	Format	Price / unit (approx)	Monthly therapy cost (Rs)	Monthly therapy cost (US\$)
Sun Pharma	In-house	Noveltreat	Obesity	0.25, 0.5, 1.0, 1.7, 2.4 mg weekly	Prefilled pen	Rs 900–2,000 per weekly pen	Rs 3,600–8,000	US\$38-84
		Sematrinity	T2D	0.5mg to 1 mg weekly	Multi-dose pen	Rs 750–1,300 per week	Rs 3,000–5,200	US\$32-55
Dr. Reddy's	In-house for injectable, oral peptide outsourced	Obeda	T2D (expand to obesity)	0.5mg to 1 mg weekly	Prefilled disposable pen	Rs 1,050 per dose (pen)	~Rs 4,200	US\$44
				3mg, 7mg, 14mg	Oral	Rs99/Rs135/ Rs225 per tablet	Rs 3000-6,750	US\$32-70
Zydus Lifesciences	Import	Semaglyn / Mashema / Alterme	T2D + Obesity	Multi-dose (titration allowed)	Reusable pen (cartridge)	Rs 550/week equivalent	~Rs 2,200	US\$23
Lupin	Zydus	Semanext / Livarise	T2D + Obesity	Same as Zydus	Reusable pen	Rs 312/week equivalent	Rs 1250	US\$13
Alkem	In-house	Semasize / Obesema / Hepaglide	T2D + Obesity	Standard weekly titration doses	Pen (disposable + reusable)	Rs 450/week	~Rs 1,800–3,500	US\$19-37
Natco Pharma	Import	Semanat / Semafull	T2D + Obesity	Multi-dose vial (weekly dosing)	Vial + pen	Rs 300/week	~Rs 1,290–1,750	US\$14-18
Eris	Natco	SUNDAE / Semanat / Semafull	T2D + Obesity	Same as Natco	Vial + pen	~Rs 220–300 per dose	~Rs 1,290	US\$14
Glenmark	Natco	GLIPIQ	T2D	Weekly dosing	Vial + pen	Rs 325–440/week	~Rs 1,300–1,800	US\$14-19
Torrent Pharma	DRL/Zydus	Semalix/ Sembolic	T2D + Obesity	Oral: 3 / 7 / 14 mg daily Weekly pen	Injectable + oral tablet	NA	~Rs1490-2500	US\$16-26
MSN Labs	In-house (via One-source API partner)	Semabest	T2D	Weekly pen	Prefilled pen	Rs 1122.5/week	Rs 4,490	US\$47
Emcure	Partnered with Novo Nordisk	Proviztra	T2D+Obesity	0.25 mg / 0.5 mg, 1 mg / 1.7 mg	Prefilled Pen	~Rs 1000-2000/week	~Rs 4000-8000	US\$40-80
Mankind	MSN	Semaglu	T2D + Obesity	Weekly pen	Prefilled pen	Rs 1500/week	Rs 5,999	US\$63
Eli Lilly	Inhouse	Mounjaro	T2D	2.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg and 15 mg (4 doses pen)	Prefilled pen	Rs. 2775-5800/week	~Rs. 11150- 23200	US\$117-244
Cipla	Partnered with Eli Lilly	Yurpeak	T2D	2.5 mg, 5 mg, 7.5 mg, 10 mg, 12.5 mg and 15 mg (4 doses pen)	Prefilled pen	Rs. 2953-5800/week	~Rs. 11800- 23200	US\$124-244

Source: Tata 1 mg, Morgan Stanley Research, Updated as on 5th June 2026, USD/INR Rate of 95

Exhibit 21: Share of semaglutide market in US\$m

Semaglutide market (US\$ Mn)	Brands	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Novo Nordisk	Rybelsus, Wegovy, Ozempic	3.7	4.2	4.1	4.2	4.9	4.7	4.6
Torrent	Semalix, Sembolic	0.0	0.0	0.0	0.0	0.0	0.5	1.8
Dr. Reddys Labs	Mashlo, Obeda, Olimviq	0.0	0.0	0.0	0.0	0.0	0.2	0.4
Zydus Healthcare	Alterme, Mashema, Semaglyn	0.0	0.0	0.0	0.0	0.0	0.1	0.4
Lupin	Livarise, Semanext	0.0	0.0	0.0	0.0	0.0	0.1	0.4
Sun Pharma	Noveltreat, Sematrinity	0.0	0.0	0.0	0.0	0.0	0.1	0.2
Emcure Pharma	Poviztra	0.0	0.0	0.0	0.1	0.1	0.1	NA
Alkem	Hepaglide, Obesema, Semasize	0.0	0.0	0.0	0.0	0.0	0.1	0.3
USV	Usema	0.0	0.0	0.0	0.0	0.0	0.0	0.2
Eris Ls	Bluvial, Sundae	0.0	0.0	0.0	0.0	0.0	0.1	0.3
Glenmark	Glipiq	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Natco	Semafull, Semanat	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Abbott	Extensor	0.0	0.0	0.0	0.0	0.0	0.0	NA
Others	-	0.0	0.0	0.0	0.0	0.0	0.0	0.1
MSN	Semabest	0.0	0.0	0.0	0.0	0.0	0.0	0.2
Total		3.7	4.2	4.1	4.3	5.1	6.1	9.3

Source: Pharmarack, Morgan Stanley Research, USD/INR Rate of 95

Exhibit 22: Share of semaglutide market in Rs mn

Semaglutide market (Rs Mn)	Brands	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Novo Nordisk	Rybelsus, Wegovy, Ozempic	350	400	390	400	470	450	440
Torrent	Semalix, Sembolic	0	0	0	0	0	47	170
Dr. Reddys Labs	Mashlo, Obeda, Olimviq	0	0	0	0	0	19	40
Zydus Healthcare	Alterme, Mashema, Semaglyn	0	0	0	0	0	14	40
Lupin	Livarise, Semanext	0	0	0	0	0	14	40
Sun Pharma	Noveltreat, Sematrinity	0	0	0	0	0	9	20
Emcure Pharma	Poviztra	0	0	0	10	10	8	NA
Alkem	Hepaglide, Obesema, Semasize	0	0	0	0	0	8	30
USV	Usema	0	0	0	0	0	0	20
Eris Ls	Bluvial, Sundae	0	0	0	0	0	6	30
Glenmark	Glipiq	0	0	0	0	0	2	10
Natco	Semafull, Semanat	0	0	0	0	0	2	10
Abbott	Extensior	0	0	0	0	0	1	NA
Others	-	0	0	0	0	0	0	10
MSN	Semabest	0	0	0	0	0	0	20
Total		350	400	390	410	480	580	880

Source: Pharmarack, Morgan Stanley Research

Oral semaglutide :

To date we have seen two generic launches of Rybelsus. On 21 March 2026, Torrent launched semaglutide (oral tablet) 3mg, 7mg and 14mg under brand names 'Semalix' and 'Sembolic'. April sales for the brand was Rs60mn (US\$0.6mn). On 20 May 2026, DRL was the second company to launch oral tablets under brand name Obeda, these were also similar strength tablets (3mg, 7mg and 14mg). DRL pricing was more aggressive than Torrent thus Torrent lowered prices to come closer to DRL. We expect more players to launch oral tablets in coming months. In fact, Torrent in its F4Q26 earnings call added that it expects the oral market to be 20%-30% of the injectable market.

Exhibit 23: Sema oral vs injectable market (US\$ mn)

(US\$ Mn)	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Oral Market	2.8	2.7	2.3	2.5	2.5	2.8
Innovator	2.8	2.7	2.3	2.5	2.4	2.2
Generics	0.0	0.0	0.0	0.0	0.1	0.6
Injectable	1.3	1.4	1.9	2.5	3.6	6.4
Innovator	1.3	1.4	1.9	2.5	2.3	2.4
Generics	0.0	0.0	0.0	0.0	1.3	4.0
Total Semaglutide Market	4.1	4.1	4.2	5.1	6.1	9.3

Source: Pharmarack, Morgan Stanley Research

Exhibit 24: Sema oral vs injectable market (Rs mn)

Semaglutide market in Value (division by dosage (Rs Mn))	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Oral Market	270	260	220	240	240	270
Innovator	270	260	220	240	230	210
Generics	0	0	0	0	10	60
Injectable	120	130	180	240	340	610
Innovator	120	130	180	240	220	230
Generics	0	0	0	0	120	380
Total Semaglutide Market	390	390	400	480	580	880

Source: Pharmarack, Morgan Stanley Research

Exhibit 25: Semaglutide value market share

Semaglutide Value Market share	Brands	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Novo Nordisk	Rybelsus, Wegovy, Ozempic	100%	100%	100%	98%	98%	76%	50%
Torrent	Semalix, Sembolic	0%	0%	0%	0%	0%	8%	19%
Dr. Reddys Labs	Mashlo, Obeda, Olimviq	0%	0%	0%	0%	0%	3%	5%
Zydus Healthcare	Alterme, Mashema, Semaglyn	0%	0%	0%	0%	0%	2%	5%
Lupin	Livarise, Sema-next	0%	0%	0%	0%	0%	2%	5%
Sun Pharma	Noveltreat, Sematrinity	0%	0%	0%	0%	0%	2%	2%
Emcure Pharma	Poviztra	0%	0%	0%	2%	2%	1%	NA
Alkem	Hepaglide, Obesema, Semasize	0%	0%	0%	0%	0%	1%	3%
Usv	Usema	0%	0%	0%	0%	0%	1%	2%
Eris Ls	Bluvial, Sundae	0%	0%	0%	0%	0%	1%	3%
Glenmark	Glipiq	0%	0%	0%	0%	0%	0%	1%
Natco	Semafull, Semanat	0%	0%	0%	0%	0%	0%	1%
Abbott	Extensor	0%	0%	0%	0%	0%	0%	NA
Others	Different brands	0%	0%	0%	0%	0%	3.0%	3%

Source: Pharmarack, Morgan Stanley Research

Post the launch of semaglutide generics, sema injectable volumes have seen a sharp step-up, rising from 25k units in Feb'26 to 63k in Mar'26 and 169k in Apr'26, driven largely by rapid generic ramp-up (40k in Mar to 137k in Apr). **This implies generics now account for ~80% of total sema injectable volumes, with growth being largely market-expansion led rather than pure substitution, as innovator volumes remain broadly stable (23k in Mar to 32k in Apr).** The scale of ramp has also led semaglutide to overtake tirzepatide on volume (169k vs 82k in Apr), reinforcing sema's positioning as a mass-volume product vs tirzepatide's premium positioning. Overall, early trends support a volume-led adoption phase post-LOE, with limited immediate disruption to innovator demand but clear and rapid mix-shift towards generics.

Exhibit 26: Semaglutide and Mounjaro volume

Semaglutide in Units (000)	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Semaglutide injectables (Innovator + Generics)	11	12	19	25	63	169
Innovator Injectables	11	12	19	25	23	32
Wegovy	11	10	10	11	10	17
Ozempic	0	2	8	13	12	12
Poviztra	0	0	1	1	1	2
Extensior	0	0	0	0	0	1
Semaglutide generics					40	137
Tirzepatide Injectables	72	81	89	92	77	82
Mounjaro	72	70	75	78	67	68
Yupreak	0	11	14	14	10	14

Sema oral market Units (000's)	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Total Oral market	1000	960	830	892	941	1257
Innovators	1000	960	830	892	868	789
Generics	0	0	0	0	73	468

Source: Pharmarack, Morgan Stanley Research

Supply chain dynamics in India

Scale of generic entry: Patent expiry triggered the entry of approximately 13+ generic players, with a 70–80% price drop. Generic semaglutide became available for as little as \$14 per month (approximately Rs 1,290).

Backward integration and scale: Initially most companies are importing the APIs. Few companies like DRL and Sunpharma are investing in full supply chain control, producing semaglutide API in-house and rapidly scaling production capacity. Indian CDMOs, including OneSource Specialty, are undertaking significant scale-up initiatives, targeting a ~4–5× increase in output by 2027 (e.g. Dr. Reddy's ~7mn pens in 2026 to >40mn pens by FY28 with F&F managed by OneSource). As per its FY25 Annual Report, OneSource is expanding cartridge manufacturing capacity from 40mn in FY25 to approximately 220mn by FY28.

Indian generic companies have pursued different API sourcing strategies:

- **Dr. Reddy's Laboratories (DRL):** The most notable example of full backward integration. DRL produces semaglutide API entirely in-house, with finished product manufacturing carried out by its manufacturing partner, OneSource Specialty Pharma Limited. DRL expects its semaglutide capacity to be around 6–7mn pens in CY2026, with FY28 capacity potentially exceeding 40mn.
- **Torrent Pharma:** Oral formulations are developed in-house, with API for both Semalix and Sembolic sourced from DRL. In injectables, the outsourcing model varies by product; Semalix pre-filled pens are sourced from DRL, while Symbolix pens are sourced from Zydus.
- **Sunpharma:** Sunpharma manufactures the semaglutide API on its own. The pre-filled pens for Noveltreat and Sematrinity are manufactured in Europe by leading global pharmaceutical device suppliers.
- **Lupin:** Partnered with Galenicum (a Spanish company) for finished formulations,

with Galenicum overseeing development, manufacturing, and supply, while Lupin handles regulatory submissions and commercialization across 23 countries.

- **Shilpa Medicare:** Developing both synthetic and semisynthetic API for semaglutide, with API validation batches ongoing and formulation development (both injectable and oral solid) completed.
- **MSN Laboratories:** Has vertically integrated semaglutide production, manufacturing both the API and the pen device in-house.
- **Biocon:** Describes itself as "vertically integrated" in semaglutide, supplying to Ajanta Pharma (India) for marketing in 26 countries.

Partnership with innovators

- **Cipla:** On October 23, 2025, Lilly and Cipla signed a marketing and distribution agreement for a second brand of tirzepatide in India. Cipla launched Yurpeak (tirzepatide) on December 10, 2025, making it the second brand of Lilly's tirzepatide available in the Indian market, alongside Mounjaro.
- **Emcure Pharma:** Launched Poviztra (semaglutide injection) in India on December 22, 2025, under an exclusive partnership with Novo Nordisk, becoming the first Indian company to distribute and commercialise a second semaglutide brand for weight management.
- **Abbott India:** In February 2026, Abbott and Novo Nordisk partnered to market and distribute semaglutide (globally known as Ozempic) in India, with Abbott commercialising it as Extensor to expand access to advanced diabetes care.

CDMO companies

- **Divi's Labs** (manufactures for innovator): The company currently has the largest Solid-Phase Peptide Synthesis (SPPS) scale (i.e. multiple ~3,000L reactors) in India. It manufactures fragments and amino acids for SPPS peptide drugs.
- **One Source Specialty:** The company is a pure-play specialty pharmaceutical CDMO focused on complex drug-device combination (DDC) products. It does not manufacture semaglutide API but it provides end-to-end formulation, fill-and-finish, and device assembly services to pharma partners who bring their own API.
- **Anthem Biosciences:** Anthem is the most backward-integrated semaglutide API manufacturer in India, manufacturing both the fermentation fragment (P29 precursor) and the chemical synthesis steps in-house making it one of the only companies in India to do so. This gives it a significant cost advantage to compete against Chinese API suppliers.
- **Shaily Engineering:** Shaily Engineering Plastics is a drug delivery device manufacturer which designs, develops, and manufactures pen injectors and auto-injectors for semaglutide. It does not manufacture API or formulation. Its proprietary platforms include ShailyPen Neo (spring-driven, variable and fixed dose) and Shaily Harmony.
- **Gland Pharma:** Gland fills cartridges with the API supplied by its pharma partners for three GLP-1 products namely liraglutide, semaglutide and tirzepatide

Exhibit 27: Key comments from major Indian pharma companies on semaglutide and GLP-1

Generic Companies	Concall	Comments
DRL	Q4FY26	So I believe that -- so the current landscape of basically Novo Nordisk as well as the two of us will stay there probably for several months. And then likely that others will come. I believe that the market in Canada is give or take. It's about 1/3 in -- for what we call public, 1/3 cash and 1/3 what they call private. Sure. So we believe that the -- all the prices will be let's say, in the neighborhood of, let's say, \$30 plus. Why I'm saying that number also is because in some markets, we are going to work with the partner and that reflect the net price that we have for them. Obviously, they will have their margins. And these numbers may go down if the competition intensified, but I don't envision it to -- at any case to be below \$25. We launched Noveltrat on 21st March. So I would say what reflection you are seeing both in IMS or AWAOS or even the prescription may not be adequate because any new product when we launch, it takes time for a complete reflection. So maybe in the next 5, 6 months, you will see the reflection, which is close to reality.
Sunpharma	Q4FY26	Being said that, the product has been received very well by the doctor's community. And one of the differentiating point is our auto-injector as well as the pain system, which was appreciated by the doctors and patients equally well. So current reflection, what you are saying is right. We are -- what you are saying, we are not #1 player in the GLP-1 generics. But as I said, the reflection and the reality takes some time to reflect, and our endeavor is always to remain leader in the market.
Mankind Pharma	Q4FY26	There's a mad rush in the market. Every company is launching. And if at this particular time, you launch, you will be lost somewhere. Mankind is focusing on adjacent therapies (vitamins, minerals, proteins) alongside GLP-1, believing a complete portfolio approach will provide a competitive advantage.
Natco Pharma	Q2FY26	I think -- and we are able to disrupt the market. But still, the market is in terms of sale, it's still with the pen. But the proposition on the vial is this, okay? The pen sells for, I mean, a normalized between INR 3,000 to INR 4,000. The vial sells for less than INR 1,000. So there's a market which is willing -- who's unwilling to pay INR 3,000 to INR 4,000, is willing to pay only INR 1,000 per month. So that market we're able to capture.
Lupin	Q4FY26	I think it's been a pretty solid launch, pretty much as good as expected or even better. We always expect it to be a strong player given our cardiometabolic play. The product from Zydus is a unique pen, and I think that's differentiation in the market as well. Our patient support program is pretty solid as well. So good start. I think we're the #2 company as a generic one, the #3 product as a product itself. So good start, but very early days, right? I mean we're talking about 9 days of data. You saw how quickly Mounjaro became the biggest product in India. And now it's flattened out, right? So how important is the generic? How important is Mounjaro versus semaglutide. I think there are multiple things to play out. There are so many other products that are going to come along as well. But clearly, this is a pretty widely accepted therapy. The number of prescribers is significant. The number of patients that have gone on to this is very material as well. So yes, I think this is one of the biggest launches for the industry.
Torrent Pharma	Q4FY26	I think we may be looking at something like INR 200 crores to INR 250 crores for semaglutide at this stage. It could vary. We can share an update in the next coming quarters, but that's the difference in the opportunity size that we're seeing.
Zydus Pharma	Q4FY26	During the quarter, we launched semaglutide injection under the brand names SemaglynTM, MashemaTM, and AltermeTM. in a reusable multi dose pen, enabling flexible dosing, better adherence, and lower therapy costs.
Eris Lifesciences	Q4FY26	Our Sema launch was in a phased manner post the loss of exclusivity. We launched Sundae Vial in March. Sundae Pen 2 mg came in mid-April. We launched the 4 mg Sundae Pen in early May and we expect obesity SKU to be launched in July. Our products are priced lower than competition.
Shilpa Medicare	Q4FY26	Semaglutide as a product, our main strength is we are doing both synthetic and semisynthetic API. And also, we are forward integrating with our own formulation. So semaglutide itself is a very big opportunity. And because our facility has the global accreditations, we still feel in such a big opportunity if we get even small pie of that market share, right? That is a decent revenue which is possible with the product.
Biocon	Q3FY26	We have filed in Canada, Brazil, Saudi, Turkey, and we continue to file in other markets. Of course, it's -- the review cycle is long drawn, especially in markets such as Canada, where we have not seen a single generic GLP being approved, including liraglutide, which has not been approved by Health Canada. So we are hoping that sometime next calendar year, we should be in a position where we at least make advanced progress on semaglutide.
Partnership with Innovators		
Emcure (Novo Nordisk)	Q4FY26	Strong exclusive global data and established device distinguishes Poviztra from the generics, which have been introduced. The initial uptake has been promising, especially after the price cut in April 2026. And I must say the pricing which has been done is very competitive, and we obviously have a story and strategy
Cipla (Eli Lilly)	Q4FY26	Now with respect to semaglutide generic, as and when that happens, we believe it will open up a new segment in the market because the molecules are different. Tirzepatide has dual action, GLP and GIP, which we believe makes it a preferred option. However, from a pricing perspective, once genericization of sema happens, it might open up options at a different price point to certain segments of the market, which effectively will grow the market as opposed to eating into each other
CDMO's		
Gland Pharma	Q3FY26	So -- but what we are saying is, in addition to GLP, we are also -- because we're already making insulin for Eli Lilly for several years. So we also have that experience. So we're talking to insulin manufacturers who have commercialized this product in large numbers. So that could be a big item for this line. It is for the 40 million lines. In the meanwhile, we look at how the market behaves and if we have to sign more contracts or are the contracts what we have signed is good enough to fill those banks.
Shaily Engineering	Q4FY26	I think if I look at all the top -- the first 6 filers in Canada, 70% is ours, I believe, 70% is ours. In Brazil, it's a little bit more difficult because in Brazil, it's a lot of partnership filings, not direct from our customers. So -- and we are not privy to all the partners in Brazil. But I think we will be -- I'm confident we will be in the first wave of Brazil approvals.
Anthem Biosciences	Q4FY26	With the new facility, which has just come up. So that gives a unique that gives a good proposition for customers to keep the products more developed. Development or in a late scale project. Also to come to us because you don't then need to venture out somewhere else, looking out for capacity to to tech transfer the project. So that's a differentiator for us on the generic peptide. GLP one. I did talk about we are completely backward integrated over there. We manufacture the. Fermentation fragment as well as the synthesis. And only company in India to be completely backward integrated. And hence we competitive from a cost point of view to play, to compete against the Chinese on semaglutide. So that's the uniqueness or the competitiveness. When we target our Indian customers who are looking at tapping the markets as well as India markets in the GLP one space, the generic semaglutide space side
Divis Labs	Q2FY26/ Q4FY26	Divi has strategically decided that we will not look at generic part of peptide synthesis. We are right now fully occupied with the amount of CS projects we have. So, we do not want to venture into that mode. I would say in terms of scale, I can give you a rough thing. We have several 3,000 liter SPSS, which by far in India nobody has. We are quite strong and committed towards this segment, and we are targeting to be one of the largest players -- global players in the world.
Neuland Laboratories	Q4FY26	So I think the pitch we make to peptide clients is that we've invested close to 16, 18 years in the peptide space, and we've kind of worked our way organically by making fragments, building blocks, amino -- natural amino acid-based fragments and then slowly moved up the value chain, making even peptide APIs. Our biggest strength is the fact that we have done a lot of process development for peptides in-house in Neuland. We've always had peptide R&D in Neuland for several years, focused on process chemistry, not medicinal chemistry. This is a skill set that is very important when it comes to engaging with clinical stage peptides, especially APIs. So we were able to successfully showcase those capabilities

Source: Company data, Morgan Stanley Research

Brazil Market Formation (2026): Semaglutide's LOE Sets the Stage for Formal Market Expansion

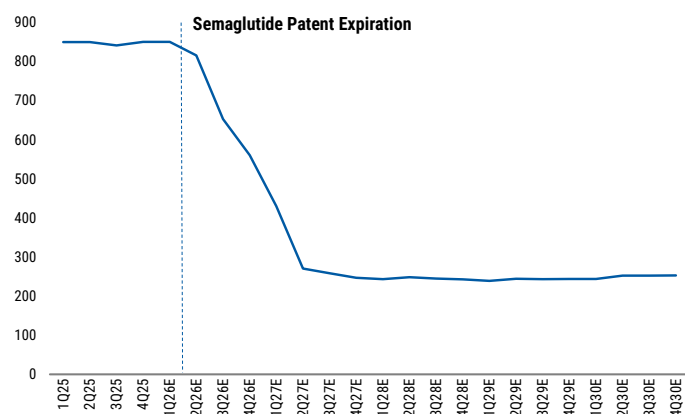
Current State Pre-Generic Entry: GLP-1 access remains constrained primarily by affordability. Brazil's GLP-1 market remains materially underpenetrated, not because of weak demand, but because affordability continues to cap adoption in an almost entirely out-of-pocket market. Monthly treatment costs for leading GLP-1 therapies still range from roughly R\$800 to R\$1,600 (US\$160 to US\$320 per month), levels that restrict access to a narrow upper-income tier and keep penetration well below the market's underlying clinical potential. The scale of demand for compounded alternatives and unauthorized imports further supports this affordability bottleneck, suggesting that many patients are actively seeking lower-cost access outside the formal market rather than opting out of treatment altogether. Supply constraints appear less broad-based today, though availability remains inconsistent for Mounjaro/tirzepatide, where recurring stockouts can still limit realized volumes even within the high-income addressable group. As a result, Brazil currently has only ~2.1mn active GLP-1 users by our 1Q26 estimates (1.3% of the current adult population), underscoring the large pool of unmet demand in a sizeable emerging market where category penetration remains well below potential.

Early pricing moves suggest incumbents are already preparing for semaglutide competition. We have also seen some initial discounts and price reductions on semaglutide's authorized generic from Eurofarma, see [LatAm Healthcare: GLP-1 Brazil Tracker: April Imports Remain Solid Despite MoM Decline; New ANVISA Data Adds Insights \(7 May 2026\)](#). These moves are still immaterial to overall market dynamics, as price points remain too high to materially broaden access or trigger a meaningful shift in category penetration. However, they are directionally important: the early discounting suggests that the Novo Nordisk-Eurofarma partnership is already preparing for incoming competition ahead of broader semaglutide generic entry, with pricing likely to become a more relevant competitive lever as additional approved alternatives reach the market.

Generic-led transition in 2026: the beginning of the price reset and accelerating adoption. Following the expected entry of semaglutide generics in 2H26, we forecast a marked expansion in GLP-1 access as affordability improves, though the market may take time beyond 2026 to reach its post-LOE equilibrium. Semaglutide generics have been delayed, with [the first approval \(for EMS product\) granted in May 2026](#) with a launch planned for June 15th; we are therefore modeling market formation in 2H26; see [LatAm Healthcare & Education: Our takes from Brazil Journal HC Conference \(28 Apr 2026\)](#). Once the first branded generics launch, semaglutide prices should begin to step down, with our forecast for semaglutide generics reaching ~R\$687/month (US\$137/month) by YE26, from ~R\$1k/month (US\$200/month) currently for Novo's offering (Ozempic and Wegovy). We estimate total GLP-1 market sales of ~R\$14.3bn (US\$2.5bn) in 2026, with active users rising to ~3.4mn by YE26, as lower entry prices allow volumes to better reflect underlying demand. That said, 2026 should still be viewed as an acceleration year rather than a steady state, with additional entrants, ongoing price compression, and broader affordability gains likely to play out gradually through late-2026, leaving penetration still below long-term potential by year-end.

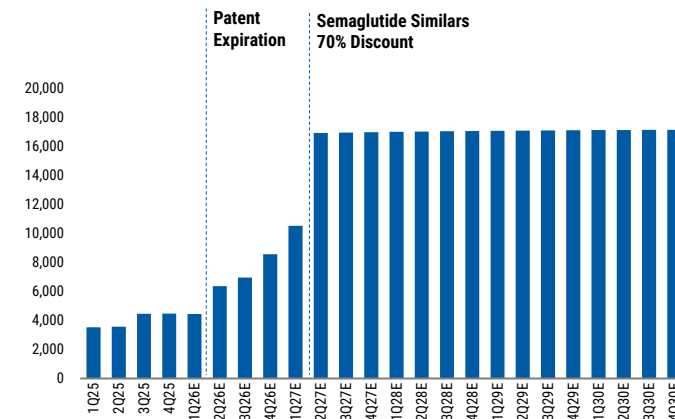
Post-LOE Shift: semaglutide generics unlock a step-change in access and market stabilization. Beyond the initial 2026 adjustment, semaglutide's loss of exclusivity may drive a structural shift in Brazil's GLP-1 market, as lower prices unlock a significantly broader addressable population in a country with steep income distribution. We expect the entry of multiple generics to result in a cumulative ~70% reduction in monthly treatment costs, with semaglutide reaching R\$294/month (US\$59/month) post-LOE (see [Exhibit 28](#)). At this price point, affordability expands meaningfully into middle-income households, shifting GLP-1 access from a high-income niche toward a broader mass-market therapy. We estimate that the addressable GLP-1 population expands to ~17mn potential users post-LOE (see [Exhibit 29](#)), with Mounjaro and future Lilly offerings likely concentrated in higher-income cohorts, while semaglutide becomes the preferred option across Brazil's large B2-C income classes. As adoption broadens, we expect the GLP-1 market to continue scaling after 2026, reaching ~R\$20.6bn (US\$3.4bn) in 2027, ~R\$30.9bn (US\$4.3bn) in 2029, and stabilizing around a long-term opportunity of ~R\$39bn (US\$5.1bn), driven primarily by price elasticity and stronger consumer willingness to pay, rather than changes in underlying obesity prevalence (see [Exhibit 30](#)).

Exhibit 28: Semaglutide average price evolution (BRL/Units)



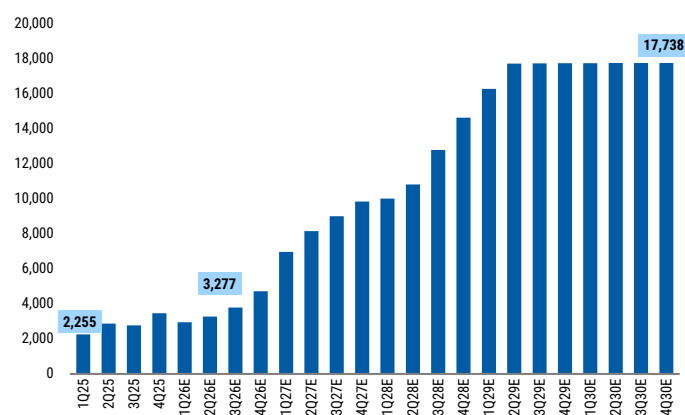
Source: IQVIA, Morgan Stanley Research estimates (E)

Exhibit 29: GLP-1 potential clientele (ex-treated patients; '000)



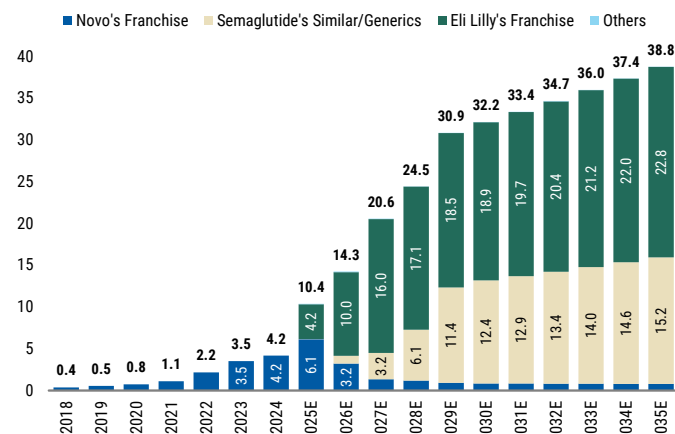
Source: IQVIA, Nexo Journal (Monthly Wage Ranking), IBGE, and Morgan Stanley Research estimates (E)

Exhibit 30: GLP-1 volumes evolution (Units; '000)



Source: IQVIA, Nexo Journal (Monthly Wage Ranking), IBGE, Morgan Stanley Research estimates (E)

Exhibit 31: GLP-1 market (BRLbn; PPP)



Source: IQVIA, Nexo Journal (Monthly Wage Ranking), IBGE, Morgan Stanley Research estimates (E)

Local producer likely to be the main beneficiary of LOE. The post-LOE semaglutide market should be shaped by partnerships between global generic producers, mostly from

India and China, and local players. Within this landscape, EMS, a non-listed family-owned company, stands out as the only domestic producer and likely the best positioned to benefit from semaglutide's LOE, with its synthetic semaglutide pen, Ozivy, becoming the first to receive Anvisa approval on May 26 (see [here](#), Portuguese only). The company has invested in local GLP-1 manufacturing capacity, with reported annual capacity of approximately 20mn injectable pens for liraglutide and semaglutide. In our view, this local footprint could give EMS a cost advantage, as it does not rely on royalty-bearing partnerships with foreign manufacturers. The Brazilian government's decision to reduce the import tariff on components used in EMS's injectable pens from 14.4% to 0% for one year on March 26, with a quota of up to 30mn units, should further support competitive pricing as production scales. Following approval, CMED set the ceiling price at levels comparable to Ozempic; however, EMS announced a retail price of BRL 452, with the potential to reach BRL 287 (US\$57) for a three-month period under the company's discount program (see [here](#), Portuguese only). Looking ahead, should EMS successfully scale production, it could broaden access through formal channels and position itself as the leading low-cost producer in the post-LOE semaglutide segment.

Potential new entrants: a broader competitive pipeline beyond EMS. While EMS is the first mover following Ozivy's Anvisa approval, the post-LOE semaglutide market should become increasingly crowded as additional players move through the regulatory pipeline. According to G1 (see [here](#), portuguese only), Anvisa had 17 semaglutide-related filings in progress, including 8 already under analysis, of which 3 were in an advanced stage awaiting company responses to Anvisa's technical questions. With EMS now approved, we believe one of these advanced files was likely Ozivy, but the broader pipeline remains meaningful. Among those names, public reporting points to potential contenders including Hypera, Biommm, through its agreement with Biocon, as well as Cimed, Prati-Donaduzzi (see [here](#), portuguese only) and potentially Sandoz, which has reportedly identified Brazil as one of its early 2026 target markets for generic semaglutide (see [here](#)). In our view, EMS may capture the initial first-mover advantage, but the medium-term market structure should evolve toward a multi-player branded semaglutide-copy landscape, with competition driven by approval timing, CMED pricing, pen supply, local distribution reach, physician relationships and the ability to fund patient discount programs.

Implications for Novo Nordisk post-LOE: Brazil as an early test case for ex-US erosion. Brazil is one of Novo Nordisk's first large emerging-market tests of post-LOE semaglutide dynamics, and the recent approval of EMS's Ozivy suggests the generic approval cycle is now beginning to move. While the delayed approval process gave Novo more time to defend its franchise, the structural risk is now becoming more tangible: as approved synthetic semaglutide products reach the market, Novo is likely to face price erosion, volume share dilution and increased competition from local players with strong distribution, prescribing relationships and retail reach. The key debate is how much of the expanding market Novo can retain as semaglutide access broadens from a premium niche toward a more mass-market therapy. In our view, Brazil should be watched as an early indicator of Novo's broader ex-US LOE headwinds, particularly in markets where affordability constraints are high, local generic capabilities are strong and demand is already visible through informal and compounding-pharmacy channels.

Impacts on our coverage: Semaglutide's LOE should be positive for both RADL (OW) and HYPE (OW), though through different mechanisms. For RADL, the main upside is

the expansion of the addressable market as lower semaglutide prices move GLP-1 access beyond a narrow high-income niche, alongside the migration of demand currently served by unauthorized imports and compounded alternatives into the formal pharmacy channel. While lower prices may pressure average ticket, elasticity should support higher volumes, traffic and recurring prescription demand, with RADL well positioned to capture this shift given its national footprint, digital capabilities, supplier relationships and ability to sell both branded and generic GLP-1s. For HYPE, the opportunity is more execution-dependent, but potentially meaningful: if the company secures timely Anvisa approval and launches in the early waves of branded semaglutide copies, it could participate directly in the expansion and formalization of a market where demand already exists but is still partly priced out of the legal channel. EMS's first-mover advantage and local manufacturing footprint are relevant competitive hurdles, but the market should evolve into a multi-player branded-copy segment, where HYPE's physician relationships, marketing capabilities, retail reach and brand-building capacity could support share gains as semaglutide shifts from a premium niche into a broader mass-market therapy.

Risks to our estimates: unauthorized imports and compounded GLP-1s highlight unmet demand, but also regulatory and safety risk. The emergence of compounded, informal, and cross-border GLP-1 channels is another signal that underlying demand exceeds the formal market's current affordability curve. Before generic entry, there was an expectation that lower-cost alternatives would reduce average prices and expand access, but the size of the parallel market suggests that demand among lower-income consumers already exists, even before formal semaglutide generics have arrived. For reference, according to RADL's 1Q26 conference call, **the unofficial GLP-1 market could be around 3-4x the size of the official market.** Going forward, a key catalyst would be tighter enforcement on unauthorized imports and compounded alternatives, which could pressure black-market supply, while lower prices from approved generics should encourage consumers to transition into the legal market.

China market formation (2027)

Although China's GLP-1 market is much smaller than the US, it possesses significant peptide API capacity, the result of a serendipitous set of conditions.

China's current GLP-1 market is small compared to the US market: In 2025, the total GLP-1 market in China was only Rmb8.1bn (~US\$1.1bn), growing 27% YoY. Close to 90% of the market came from semaglutide (IQVIA), a fraction of the US market in 2025 (US\$52bn [also from IQVIA], up 37% YoY). Part of the explanation lies in later approval dates for the branded drugs and a low level of reimbursement status. Of the 10 GLP-1 brands launched since 2010 in China, most launches lagged the US by several years including Ozempic (3.5 years) and Wegovy (3 years), with Mounjaro launched for diabetes 2 years after the US.

Exhibit 32: GLP-1s approvals in China have lagged the US

Brand	Generic name	Company	Indication	China approval date	US approval date	China's lag
Victoza	Liraglutide injection	Novo Nordisk	Diabetes	March, 2011	January, 2010	~1 year
Bydureon	Exenatide ER injection	AstraZeneca	Diabetes	January, 2018	January, 2012	~6 years
Trulicity	Dulaglutide injection	Eli Lilly	Diabetes	February, 2019	September, 2014	~4.5 years
Ozempic	Semaglutide injection	Novo Nordisk	Diabetes	April, 2021	December, 2017	~3.5 years
Saxenda	Liraglutide injection	Novo Nordisk	Obesity	July, 2023	December, 2014	~8.5 years
Rybelsus	Semaglutide tablet	Novo Nordisk	Diabetes	January, 2024	September, 2019	~4.5 years
Mounjaro	Tirzepatide injection	Eli Lilly	Diabetes	May, 2024	May, 2022	~2 years
Wegovy	Semaglutide injection	Novo Nordisk	Obesity	June, 2024	June, 2021	~3 years
Mounjaro/Zepbound	Tirzepatide injection	Eli Lilly	Obesity	July, 2024	November, 2023	~0.5 year
Xinermei	Mazdutide injection	Innovent	Obesity	June, 2025	NA	NA
Xinermei	Mazdutide injection	Innovent	Diabetes	September, 2025	NA	NA

Note: Tirzepatide for obesity is marketed as Zepbound in the US, and as Mounjaro in China for both diabetes and obesity. Source: NMPA; US FDA; Morgan Stanley Research.

China is largely a single payer (government) market; with reimbursed status key to driving formulary inclusion and uptake in public hospitals. According to an estimate by biotech company Innogen, GLP-1 penetration in China remains low compared to the US/EU with <10% penetration for diabetes and negligible for obesity. A recent survey by our US team indicated a number of barriers to patient uptake - side effects, high cost, and needle phobia ([see here](#)). Another key factor explaining the small market size by value is that a number of these GLP-1s have gone generic/biosimilar in China. Victoza/Saxenda's compound (liraglutide) expired in 2017, leading to generic launches by Huadong, Tonghua Dongbao, and Sino Biopharma in 2023-2024. Luye also launched a Trulicity (dulaglutide) biosimilar in August 2025. Public GLP-1 reimbursement status in China only covers the diabetes indication, with obesity not reimbursed.

Exhibit 33: GLP-1 reimbursement status in China - only the diabetes indication is reimbursed

Brand	Coverage by public insurance	NRDL inclusion date	Indication
Victoza	Yes	2018-2019	Diabetes
Bydureon	No	NA	NA
Trulicity	Yes	January, 2022	Diabetes
Ozempic	Yes	January, 2022	Diabetes
Saxenda	No	NA	NA
Rybelsus	No	NA	NA
Mounjaro	Yes	January, 2026	Diabetes
Wegovy	No	NA	NA
Zepbound	No	NA	NA
Xinermei	No	NA	NA

Source: NMPA; Morgan Stanley Research

The grey/unofficial channel in China is significant. This market consists of online sales, counterfeit products, and sales of raw APIs. Chinese authorities have intensified crackdowns on unlicensed production and sales of GLP-1s in recent months; for example, beginning mid-May, pharmacies require hospital paper prescriptions from customers in order to dispense GLP-1 drugs. Although the official market in China is small, a significant portion of the addressable market has so far leaked through unofficial channels.

New launches from Lilly, supported by local CDMOs, could help drive market growth before generic entry. Tirzepatide (Mounjaro) was just included into the NRDL earlier this year, so volume would likely expand this year (with a price concession). Upcoming launch of Lilly's oral GLP-1 Foundayo (orforglipron), could also drive uptake with a differentiated convenience. In early 2026, Lilly partnered with Chinese CDMO Pharmaron to supply Foundayo ([note](#)); we note that Pharmaron's oral solids facility in Beijing has an annual capacity of 1-2bn tablets. The site could potentially be dedicated to the drug, which suggests significant market potential ([note](#)). Leading Chinese CDMO WuXi AppTec has also been a long-standing global partner of Lilly's.

The patent for semaglutide expired in China in March 2026. The compound patent for semaglutide was filed in March 2006; and was set to expire in China March 2026 under the 20-year patent term set by the WTO, earlier than in the US and EU where Novo leveraged specific patent term extension mechanisms. No such mechanism existed in China prior to June 2021, when the government introduced a similar extension mechanism modeled on US/EU's. Semaglutide (Ozempic) was first approved in China in April, 2021 for the diabetes indication, thus narrowly missing the enactment of the extension pathway.

However, the exclusivity is extended potentially until April 2027 under the China-Switzerland Free Trade Agreement (CSFTA). Novo received approval for semaglutide in April 2021 in China through a filing was via its Swiss-registered entity, Novo Nordisk Pharma AG. In 2014, China and Switzerland signed the CSFTA, a bilateral economic treaty that reduces trade barriers, improves market access, and strengthens IP protection. For the latter, the two countries grant six years of market protection for undisclosed pharmaceutical test data, counting from the drug's approval date (i.e. April 2021 for semaglutide in China). When the patent expired in March 2026, Novo invoked this specific clause, barring generic launches in China until April 2027. Due to this clause, generic applicants cannot use Novo's trial data in their regulatory submissions till then. Hangzhou

Jiuyuan challenged the CSFTA in the Zhejiang Provincial court in January 2026; but at time of writing, no generic has been approved in China.

In anticipation of semaglutide's patent expiry, a host of Chinese developers have geared up applications for a biosimilar. These include major pharma companies like Fosun and Sino Biopharm (Chia Tai), and specialty companies (either in diabetes or in insulins/peptides) like United Labs and Huadong. Most applications in the queue are for the diabetes indication. One of the frontrunners in this race, Huadong, indicated at its end-May shareholders meeting that because of CSFTA, its semaglutide generic would not launch until after April 2027, thus confirming the validity of the extension. Meanwhile, Hybio has also recently submitted a biosimilar application (co-developed with CR Double-Crane) and is expected to serve as the API supplier.

Exhibit 34: Semaglutide biosimilars accepted for review by NPMA in China

Date of Acceptance by NMPA	Company	Indication	Registration Pathway
Apr-24	Jiuyuan Genetic (Jiyoutai)	Type 2 Diabetes	3.3
Jun-24	Xinbeijiang Pharma	Type 2 Diabetes	3.3
Sep-24	Qilu Pharma	Type 2 Diabetes	2.2
Jan-25	The United Laboratories	Type 2 Diabetes	3.3
Apr-25	Zhongmei Huadong	Type 2 Diabetes	3.3
Aug-25	BCHT Biotech	Type 2 Diabetes	2.2
Aug-25	Huisheng Biopharma	Type 2 Diabetes	3.3
Sep-25	Beite Biopharma	Type 2 Diabetes	3.3
Nov-25	Fosun Wanbang	Type 2 Diabetes	3.3
Dec-25	BCHT Biotech	Obesity and Overweight	2.2
Dec-25	Chia Tai Tianqing	Type 2 Diabetes	3.3
Feb-26	Jiuyuan Genetic (Jikeqin)	Obesity and Overweight	3.3
Apr-26	Beite Biopharma	Obesity and Overweight	3.3
Apr-26	Zhongmei Huadong Pharma	Obesity and Overweight	3.3
May-26	CR Double Crane/ Hybio Pharma	Type 2 Diabetes	2.2

Source: NMPA; Morgan Stanley Research

LLY Considerations

LLY injectable incretin supply outlook

At peak, we estimate LLY's current injectable platform (autoinjectors + KwikPen) can deliver ~\$100bn in total sales capacity, based on our latest assumptions of \$360/\$195 revenue per script in the US/OUS, respectively, while serving the equivalent of ~25mn patients globally ([Exhibit 35](#)). Vials offer another supply node (but we have limited visibility on capacity here), and LLY could elect to make future investments to build additional injectable capacity as well. That said, we believe that in order to achieve statin-like penetration levels of the US market for GLP-1's, orals (such as Foundayo) will likely be needed given injectable capacity constraints (see [HERE](#)).

Exhibit 35: Estimated peak sales from LLY's incretin injector platform (TRx/people in 000's)

LLY peak capacity (MSe)	Annual TRx	# of people	\$/script	Sales (\$bn)
Auto-injectors				
Base	30,000	2,500	\$360	\$11
RTP, NC (Aug 2023)	30,000	2,500		\$11
National Resilience, OH (2024)	50,000	4,167		\$18
Concord, NC (4Q24/1Q25)	30,000	2,500		\$11
Pleasant Prairie, WI (YE25) - Part I	10,000	833		\$4
Pleasant Prairie, WI (YE28) - Part II	15,000	1,250		\$5
BSP, Rome (YE25)	15,000	1,250		\$5
Alzey, Germany (1H27)	30,000	2,500		\$11
Lehigh Valley, PA (~2031)	37,500	850		\$14
Total auto-injectors (excl. PA)	210,000	17,500		\$89
Kwikpens	55,000	4,583	\$195	\$11
Total	265,000	22,083		\$100

Source: Morgan Stanley Research; Note: US pricing assumptions are based on peak sales revenues for the combined Mounjaro/Zepbound franchise (2031 in our model). We assume an approximately 50%/50% revenue split between the two products to derive the average autoinjector pricing used in the exhibit.

We have previously estimated (see below and our prior analysis [HERE](#)) that LLY's capacity during 3Q22 was ~30mn autoinjector TRx, and reached ~90mn from bringing the two additional North Carolina facilities online. LLY has also announced that it is beginning construction on another new facility in Germany, which it expects be online in 2027 (which could be a similar footprint to RTP/Concord). LLY acquired an injectable manufacturing facility from Nexus Pharmaceuticals (privately held, [LINK](#)) in Prairie, WI, which could add an incremental ~10mn annual TRx to the company's existing footprint based on our work - see [HERE](#). Further, on December 5, 2024 LLY announced ([LINK](#)) a \$3bn expansion of their manufacturing facility in Wisconsin. Per the PR, the expansion will increase LLY's injectable manufacturing capabilities to meet demand for diabetes/obesity drugs, as well as future pipeline products. Additionally, on January 30, 2026 LLY disclosed plans to build a new manufacturing facility in Lehigh Valley, Pennsylvania for \$3.5bn ([LINK](#)). Management indicated the site will serve as the company's newest injectable medicine and device manufacturing facility, although it remains unclear at this stage how much of the capacity will ultimately be allocated toward incretins vs. other medicines. Based on the disclosed total manufacturing footprint (~925K sq. feet; [LINK](#)), we estimate the parenteral production capacity may ultimately be modestly larger than RTP (~750K sq. feet - [LINK](#))/Concord, with the site expected to become operational in 2031. We updated our peak injectable capacity estimates, assuming an additional ~37.5mn annual TRx that begin to come online by 2031-2032; see [Exhibit 35](#).

We previously estimated that LLY could produce ~55mn KwikPens (serving ~4.6mn patients/year) by leveraging existing excess capacity (see [Exhibit 36](#) below). While we caveat that we have limited visibility into LLY's OUS volumes and LLY's amount of excess capacity, we believe our estimates demonstrate that even under more conservative assumptions than we are currently making, LLY could substantially increase its current production levels. In our view, oral medications will be a key factor in the market's growth given the supply constraints faced by injectables related to fill-and-finish for the injection devices. Further, a subset of patients generally will prefer pills over injections in any drug market, and we expect this to be the case in obesity as well.

See our prior work on LLY's supply ramp in "[Mapping GLP-1 Supply](#)".

Exhibit 36: LLY KwikPen capacity estimate

Key Assumptions	
US % of WW Volumes	45%
Capacity Utilization	70%
Kwikpen excess capacity for OUS Tirzepatide	80%

Kwikpen Volumes

US Insulin Kwikpen Volumes	72
OUS	88
WW Kwikpens Sold	160
WW Capacity @ 100% Utilization	229
Excess available capacity @ 100% Utilization	69
Excess Available Capacity for Kwikpens (mn)	55
Patient capacity (mn)	4.6
Autoinjector Peak Capacity Estimates	

LLY GLP-1 capacity analysis

Annual TRx (mns)

Mounjaro+Trulicity TRx 11/4/2022	22
Mounjaro+Trulicity TRx Peak in LTM	26
Avg Mounjaro+Trulicity US TRx - 2022	24
Trulicity TRx 11/4/22	13
Trulicity TRx Peak in LTM	15
Avg Trulicity ex-US TRx (estimated)	6
LLY Capacity as of 4Q22	31
Patient capacity (mn)	2.6

Source: Morgan Stanley Research estimates, Company data, IQVIA

On March 11, 2026, LLY indicated plans to invest \$3bn in China over the next decade to expand local supply-chain capacity, including establishing localized manufacturing for oral solid dosage forms ([LINK](#)). The investment will primarily support production of Foundayo (oral GLP-1), for which LLY China submitted marketing applications to the NMPA (China's FDA equivalent) for obesity and T2D at the end of 2025. LLY noted cumulative investments in China now total ~\$6bn across cardiometabolic health, neuroscience, oncology, and immunology. In our view, the latest investment underscores LLY's confidence in the commercial potential in China for Foundayo.

Outside the U.S., LLY initially launched Mounjaro (tirzepatide) in select markets using a syringe/vial presentation before transitioning to the KwikPen format following initial regulatory approvals in the UK and EU, per 1Q24 earnings commentary ([LINK](#) and [LINK](#)). The subsequent KwikPen rollout expanded across Germany, Saudi Arabia, and the UAE, with approval in Spain disclosed during 2Q24 EPS results commentary, followed by launch in Australia. **As of 1Q26, Mounjaro had been fully launched in more than 55 countries**, with particularly strong uptake in Brazil and Korea, where estimated market share reached ~60% (see [Exhibit 37](#) for details on the timing of select international market launches).

Exhibit 37: Select OUS regulatory approvals and launches for LLY's GLP-1 obesity franchise

Country / geography	Brand/Type	Obesity approval date	Commercial launch / availability date
United States	Mounjaro (T2D)	May 13, 2022	Week of June 6, 2022
United States	Zepbound (Obesity)	Nov. 8, 2023	Week of Nov 24, 2023
United States	Foundayo (Obesity)	Apr. 1, 2026	Week of Apr 6, 2026
UAE	Mounjaro (T2D/Obesity)	Prior to Aug. 31, 2022	Aug. 31, 2022
United Kingdom	Mounjaro	Nov. 8, 2023	Private market rollout during Q1 2024; NHS phased rollout Jun. 2025
European Union	Mounjaro	Dec. 11, 2023 (EC decision)	Varies country-by-country through 2024; no single EU-wide launch date
Switzerland	Mounjaro	Jul. 9, 2024	2H24
China	Mounjaro	Jul. 19, 2024	2024 rollout after approval
South Korea	Mounjaro	Jul. 30, 2024	Aug. 20, 2025
Australia	Mounjaro	Sep. 20, 2024	Late 2024 / early 2025 rollout
Hong Kong	Mounjaro KwikPen	Oct. 28, 2024	Available by YE'24
Japan	Zepbound	Dec. 27, 2024	Apr. 11, 2025
Thailand	Mounjaro	Approval prior to May 2025	May 2025
Canada	Zepbound KwikPen	May 13, 2025	Jul. 9, 2025
Brazil	Mounjaro	Jun. 9, 2025	2025 rollout expected after approval
Singapore	Mounjaro KwikPen	Jun. 18, 2025	Aug. 29, 2025
UAE	Foundayo	Apr 3, 2026	May 2026

Source: Company data, Morgan Stanley Research

Mounjaro 2026 OUS opportunity

We recently conducted sensitivity analysis for LLY's ex-US Mounjaro opportunity based on historical share gains and accelerating GLP-1 obesity adoption/growth trends (see [HERE](#)). Based on our analysis of ex-US GLP-1/incretin market growth, prescription trends, and regional share dynamics, we see potential upside to our ex-US Mounjaro forecasts, which already sit ahead of consensus estimates ([Exhibit 38](#)).

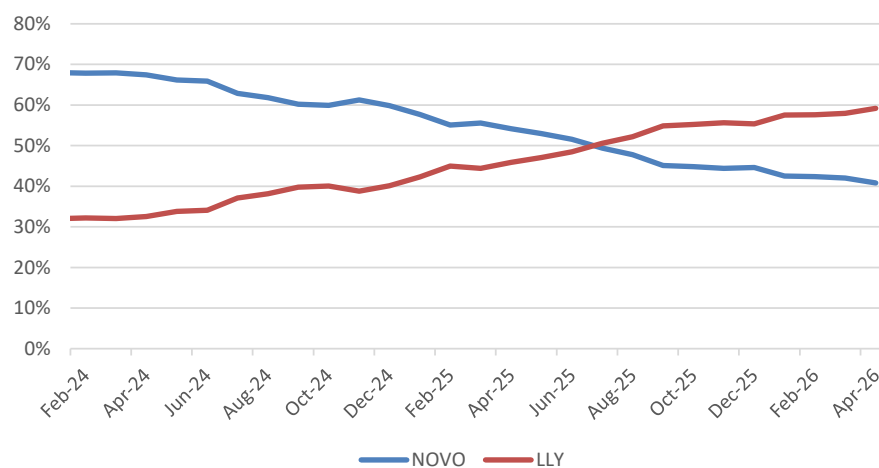
Our analysis suggests MS/cons Mounjaro OUS estimates may still prove conservative if LLY is able to sustain current international share trends, particularly as obesity adoption accelerates globally. At the same time, we acknowledge that visibility into international markets remains constrained, given limited prescription transparency outside Europe (see our EU GLP-1 script note [HERE](#)), fragmented regional pricing, uneven reimbursement frameworks, and the upcoming potential impact of generic versions of Novo's semaglutide in select markets such as India and Brazil (discussed above). As a result, while we continue to see a meaningful medium-to-long-term OUS opportunity for LLY's obesity franchise (including oral Foundayo), quarterly forecasting internationally remains inherently less precise than in the U.S.

Exhibit 38: Historical GLP-1 injectable OUS sales and LLY projected OUS sales in FY26

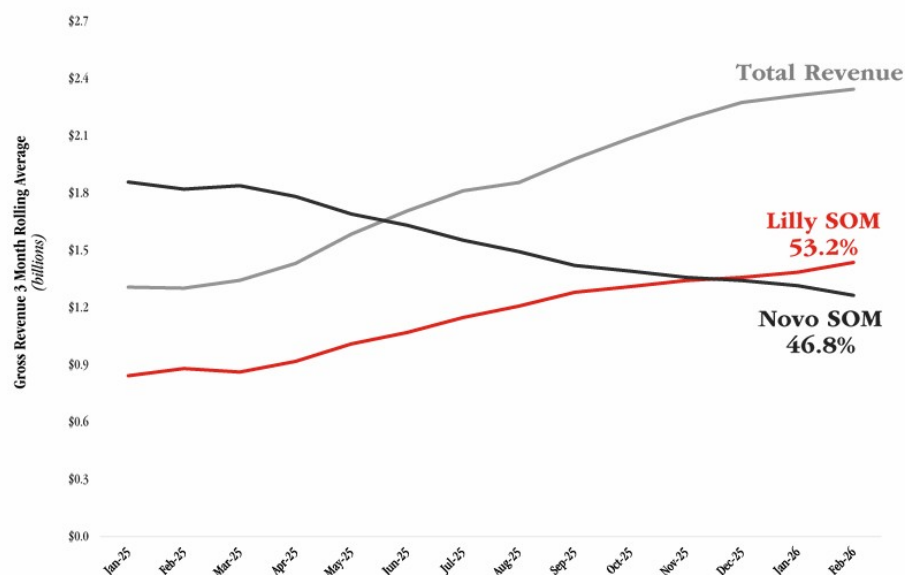
USD (\$mn)		1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E
Total Implied GLP-1 ex-US market		\$3,815	\$4,921	\$6,216	\$6,492	\$7,850			
Total GLP-1 ex-US mkt projection							\$8,751	\$9,715	\$10,679
Scenario #1 (Mounjaro as % of ex-US Total GLP-1 Sales)	2%	31%	39%	48%	50%	56%	56%	56%	56%
Mounjaro Projected ex-US sales (\$mn)						\$4,430	\$4,939	\$5,483	\$6,027
MS model						\$4,430	\$4,750	\$5,000	\$5,300
delta vs Mse						\$0	\$189	\$483	\$727
Scenario #2 (Mounjaro as % of ex-US Total GLP-1 Sales)	2%						58%	60%	62%
Mounjaro Projected ex-US sales (\$mn)							\$5,114	\$5,871	\$6,667
delta vs Mse							\$364	\$871	\$1,367
Scenario #3 (Mounjaro as % of ex-US Total GLP-1 Sales)	-2%						54%	52%	50%
Mounjaro Projected ex-US sales (\$mn)							\$4,764	\$5,094	\$5,386
delta vs Mse							\$14	\$94	\$86
VA consensus Mounjaro ex-US sales (\$mn)							\$4,553	\$4,709	\$4,892

Source: Morgan Stanley Research estimates (E); Note: linear regression was applied to total GLP-1 ex-US sales in 1Q25-1Q26 to estimate remainder of FY2026. Total implied GLP-1 market calculations include NOVO's Ozempic, Wegovy, Rybelsus and LLY's Mounjaro, Zepbound.

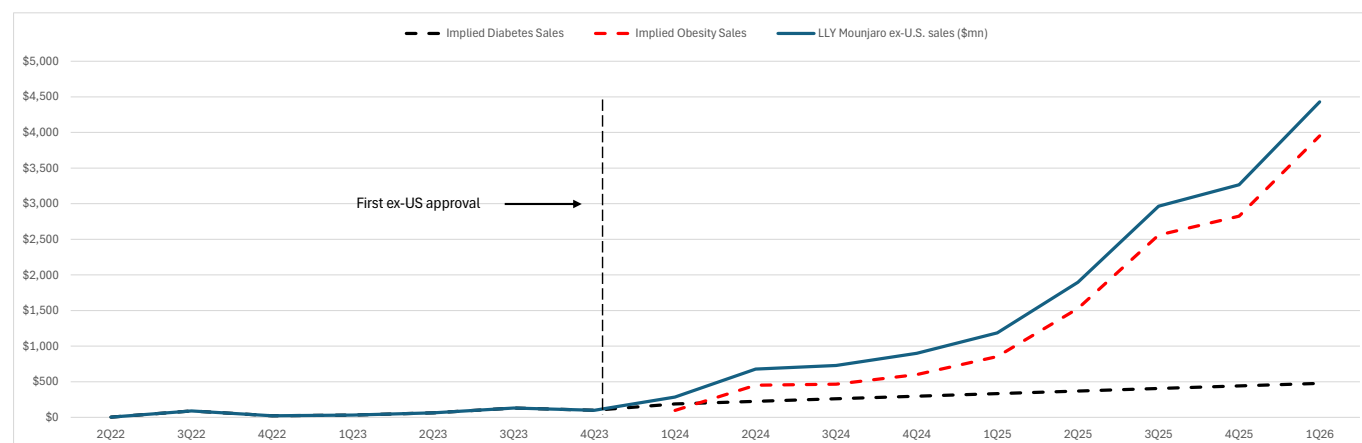
LLY noted that the company remains the global incretin market-share leader, with ~53% share internationally vs. ~60% in the U.S., and expects continued strong y/y growth with “some” sequential growth as recent market-share gains begin to stabilize. The current Mounjaro OUS mix is ~75% weight management (largely cash-pay) and 25% T2D (primarily reimbursed). Management also highlighted encouraging recent trends in India, where the launch of generic semaglutide has expanded category awareness and driven a “halo effect,” with Mounjaro reportedly growing ~10% despite generic competition. In China, Mounjaro was added to the NRDL effective January 1, creating a near-term pricing headwind but potentially supporting stronger volume growth later in 2026 ([LINK](#)).

Exhibit 39: NOVO vs. LLY value share of weekly injectable GLP-1 in Europe

Source: IQVIA Global MIDAS ([LINK](#))

Exhibit 40: International incretin analog market per LLY

Source: LLY 1Q26 EPS presentation; Note: Data source: IQVIA GMI Agent AI, MIDAS Monthly accessed 4/29/2026; US MNF: Sales in US dollars at the exchange rate in effect at the time of sale; MNF: Wholesaler purchase price and manufacturer's selling price; Market basket: Bydureon/Byetta, Lyxumia, Victoza, Ozempic, Trulicity, Rybelsus, Saxenda, Wegovy, Mounjaro, Zepbound; Geography: Worldwide excluding USA and Puerto Rico.

Exhibit 41: LLY Mounjaro OUS sales with implied obesity/T2D split (post-first OUS obesity approval)

Source: Morgan Stanley Research; Note: T2D sales estimated using a linear regression based on 4Q22–3Q23 sales, i.e., prior to the first OUS approval of Mounjaro for obesity.

Exhibit 42: LLY vs. Novo GLP-1 OUS sales and LLY market share

Mounjaro ex-US Sales (\$mn)	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26
NOVO GLP-1 ex-U.S. (\$mn)	\$747	\$808	\$888	\$987	\$1,272	\$1,614	\$1,597	\$1,662	\$1,847	\$2,047	\$2,441	\$2,523	\$2,629	\$3,024	\$3,251	\$3,225	\$3,420
LLY GLP-1 ex-US sales (\$mn)	\$3	\$90	\$23	\$32	\$64	\$132	\$101	\$286	\$677	\$728	\$899	\$1,186	\$1,897	\$2,965	\$3,267	\$4,430	\$4,430
Total GLP-1 ex-US sales (\$mn)	\$747	\$811	\$978	\$1,010	\$1,304	\$1,678	\$1,729	\$1,762	\$2,134	\$2,725	\$3,169	\$3,422	\$3,815	\$4,921	\$6,216	\$6,492	\$7,850
LLY																	
EU mkt Share (post-obesity launch)		0%	0%	0%	0%	0%	0%	0%	15%	20%	25%	30%	35%	40%	45%	50%	55%
Mounjaro (as % of ex-US Total GLP-1 Sales)	0%	0%	9%	2%	2%	4%	8%	6%	13%	25%	23%	26%	31%	39%	48%	50%	56%

Source: Company data, Morgan Stanley Research; Note: 1Q24 denotes the first launch of LLY's obesity indication (except in the UAE, where the product was used at launch in tandem with the T2D indication). See [Exhibit 35](#) for additional details. A 5% linear growth rate is assumed for EU market share proxy calculations, applied to a T2D baseline share starting in 1Q24 to estimate obesity growth. Mounjaro, expressed as a percentage of total ex-U.S. GLP-1 sales, reflects the combination of NOVO sales and LLY ex-U.S. sales.

The entry of orals

We recently updated our views on the evolving GLP-1 landscape across obesity and T2D, refreshing our global obesity model and latest GLP-1 assumptions (see ["Tracking the GLP-1 'Unlock' HERE](#)). We continue to see sustained uptake of GLP-1 therapies and

believe five key drivers can support further global penetration over time: (1) the launch of oral GLP-1 therapies (Novo's Wegovy pill and LLY's Foundayo), (2) broader access to all GLP-1's in the US for elderly patients via Medicare (at a co-pay of \$50/month), (3) international market expansion (such as China and Brazil, which we discuss in detail), (4) next-generation molecules/pathways, and (5) clinical benefits in new disease areas such as inflammation and neurological indications.

We believe orals, such as Novo's oral Wegovy pill (oral peptide GLP-1, approved on December 22, 2025 for obesity) and LLY's Foundayo (oral small molecule GLP-1, approved on April 1, 2026 for obesity) will also drive the next inflection in the obesity market. On 4Q25 EPS LLY reiterated its view that the entry of oral GLP-1's will expand the market, which was also supported by Novo's 4Q25 earnings call commentary that most oral Wegovy patients are incretin naïve. In our recent AlphaWise consumer survey, 77% of respondents on the Wegovy pill are new to GLP-1 therapy, further supporting our view that orals will substantially expand the category ([LINK](#)). On 1Q26 LLY noted that the significant inventory build speaks to upcoming international roll out of the drug as they see a "huge" opportunity for Foundayo around the world, also consistent with our view ([LINK](#)).

Oral GLP-1 therapies in development aim to deliver the benefits of existing injectable agents while offering patients an alternative formulation and, in some cases, improved scalability on a global basis. A key scientific challenge is that most incretins are peptides that are rapidly degraded in the stomach, leading companies to pursue either (1) engineering oral peptide delivery systems, such as Novo's SNAC delivery platform utilized in Rybelsus/Ozempic pill and Wegovy pill, or (2) non-peptide small molecules, such as LLY's Foundayo. Oral peptide formulations typically require food and water restrictions (e.g., administration on an empty stomach followed by a waiting period before eating, drinking, or taking other medications) to facilitate absorption. Taken together with manufacturing considerations, we believe non-peptide small molecules may ultimately prove both more scalable and more convenient. As a result, we see LLY's Foundayo playing an important role in OUS markets over time, while noting that Novo's oral Wegovy may face drug-substance scaling constraints as demand expands.

We model Foundayo 2026/2035 sales of \$1.4bn/\$29bn vs. VA cons' of \$1.3bn/\$27bn.

NOVO commentary on Oral Wegovy OUS launches

June 3, 2026 (NOVO press release): announced the launch of the Wegovy Pill in the United Arab Emirates (UAE), marking the first launch outside the United States for the obesity indication ([LINK](#)).

- Per the announcement, NOVO reiterated plans to launch the Wegovy pill in select additional markets during 2H26.

May 22, 2026 (NOVO press release): Wegovy pill (oral semaglutide) recommended by CHMP for approval in the EU as the first oral GLP-1 representing the most efficacious oral obesity treatment in its class - [LINK](#).

- "The demand for effective and convenient obesity treatment is already evident in the U.S., where more than 1 million Americans began using Wegovy® pill within the

first four months following launch. **We look forward to bringing this innovative medicine to the first markets outside the US in the second half of 2026.**"

1Q26 EPS: "...on supply of oral Wegovy as well. You've referenced launches in a couple of International Operations markets. How many markets do you think you can sustain? And generally, how is supply going?"

- "On supply, the way I'll put it is, while we do not have unlimited supply for the Wegovy pill due to the product design, then **despite the fact that we are setting a new record in terms of product uptake in the US market** with the Wegovy tablet, **we're still able to announce launches in the first markets ex-US already this year.** So, this this speaks really to scaling of supply and the inventories we have put in place. And then, **we play it gradually from here in terms of the pace of international rollout.**"
- Wegovy pill expected to launch in select markets in H2 2026.

Annual General Meeting: "the phenomenal uptake that we have seen so far and continuing. Well, play for us because we will have tens of thousands of people, if not hundreds of thousands of people by the time competition comes. That will be the testament to how they have been satisfied. And that also will help us, of course, make the pill successful. Now, we will not belittle the competition. We will prepare and will make sure that we are ready. But I think the product is helping us here. And as we do that, **we also currently are looking into ex-US, trying to find out where and how should be our next launches.** Stay tuned. You will hear of that as we will make it public."

4Q25 EPS: "Given the strong launch of Wegovy pill, is there any risk of supply outages in 2026? For example, if adoption persists at this current high level, could Novo service that demand through the end of the year with its current capacity?"

- "Over the last period, a number of occasions, I have spoken to how confident we are with regards to the Wegovy pill supply. We basically have said that we launched the pill in US at a time where we were being confident enough to know we will not run into a supply situation anymore. We have seen an incredible uptake, I would say, in the first month. And today, I will reaffirm to you that **we feel incredibly confident that we will be able to supply the US market.**"

Generic GLP-1s worth watching for Drug Packaging names

We view the launch of GLP-1 generics as incremental growth opportunities for drug packaging names such as WST and STVN. Both WST and STVN have a strong presence in GLP-1 injectables. Notably:

- GLP-1 represents approximately 18% of **WST's** total company revenue, with ~10% in elastomer components and ~8% in West Vantage (contract manufacturing). WST's portfolio participates across multiple modalities with vials, prefilled syringes, auto-injectors, and pens using WST's elastomer components. Approximately 90% of WST's GLP-1 exposure is from branded products and 10% from generics.
- GLP-1 represents 19-20% of **STVN's** total company revenue. Key products within STVN's portfolio serving the GLP market include the company's Nexa syringes and EZ-fill cartridges for multi-dose pens.

Minimal trade down expected from GLP-1 generics. Both WST and STVN primarily serve the GLP-1 market through products in their high value portfolios. Compared to products in their standard portfolios, high value products come with higher ASPs and better margin profiles. Per our conversations with management, both WST and STVN expect minimal trade down from branded GLP-1s to generics. We see generic manufacturers adopting similar high value primary packaging specifications used by branded originators in their Drug Master Files to reduce regulatory risk, allowing WST and STVN to continue supplying their most profitable products from their high value portfolio, and allowing for meaningful top line flow through to the EPS line.

GLP-1 generics could provide ASP tailwind from lesser volume discounts. Entrenched in their business models, both WST and STVN generally provide volume discounts to large customers. We believe this dynamic could be a partial explanation for why GLP-1 growth assumptions embedded in both WST and STVN's 2026 guidance come in significantly below street estimates for current GLP-1 injectables on the market. While the branded GLP-1 market is currently a duopoly, we expect the generics GLP-1 market to be more fragmented and players will not be able to receive the same level of volume discounts.

We attempt to quantify the impact of GLP-1 generics on WST and STVN's medium-term growth. Our scenario analysis assumes GLP-1 generics will be largely incremental to GLP-1 volume growth and enable expansion of the GLP-1 patient pool. Our methodology is as follows:

- We assume both WST and STVN's GLP-1 revenue growth will be tied to patient volume growth for GLP-1 injectables (as opposed to Pharma revenue growth for GLP-1 injectables on the market), which we leverage from our MS Pharma team's GLP-1 market model.
- For both WST and STVN we model a Base Case scenario with 0% GLP patient pool expansion from generic GLP-1 as a baseline reference.
- We assume two scenarios where generic GLP-1s expand GLP patient pool volumes by 10% and 20%.
- We model the ex-GLP-1 business for both WST and STVN at a constant growth rate across the three aforementioned scenarios in order to isolate the effect of

GLP-1 generics on company growth.

Exhibit 43: WST GLP-1 Generics Scenario Analysis

Base Case	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 Elastomers (\$mn)	263	316	395	474	545	599	17.9%
GLP Elastomer Growth		20.0%	25.0%	20.0%	15.0%	10.0%	
0% Patient Pool Expansion from Generics		0	0	0	0	0	
Total GLP-1 Elastomer Revenue	263	316	395	474	545	599	17.9%
Remainder of WST (\$mn)	2,811	3,005	3,185	3,376	3,579	3,794	6.2%
Remainder Growth	2.8%	6.9%	6.0%	6.0%	6.0%	6.0%	
Total (\$mn)	3,074	3,321	3,580	3,850	4,124	4,393	7.4%
Scenario #1	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 Elastomers (\$mn)	263	316	395	474	545	599	17.9%
GLP Elastomer Growth		20.0%	25.0%	20.0%	15.0%	10.0%	
10% Patient Pool Expansion from Generics		26	32	39	47	54	
Total GLP-1 Elastomer Revenue	263	342	426	513	592	654	20.0%
Remainder of WST (\$mn)	2,811	3,005	3,185	3,376	3,579	3,794	6.2%
Remainder Growth	2.8%	6.9%	6.0%	6.0%	6.0%	6.0%	
Total (\$mn)	3,074	3,347	3,612	3,889	4,171	4,447	7.7%
Scenario #2	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 Elastomers (\$mn)	263	316	395	474	545	599	17.9%
GLP Elastomer Growth		20.0%	25.0%	20.0%	15.0%	10.0%	
20% Patient Pool Expansion from Generics		53	63	79	95	109	
Total GLP-1 Elastomer Revenue	263	368	458	552	639	708	21.9%
Remainder of WST (\$mn)	2,811	3,005	3,185	3,376	3,579	3,794	6.2%
Remainder Growth	2.8%	6.9%	6.0%	6.0%	6.0%	6.0%	
Total (\$mn)	3,074	3,373	3,643	3,929	4,218	4,502	7.9%

Source: Morgan Stanley Research estimates

Exhibit 44: STVN GLP-1 Generics Scenario Analysis

Base Case	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 (€mn)	233	280	350	420	483	531	17.9%
GLP Revenue Growth	51.7%	20.0%	25.0%	20.0%	15.0%	10.0%	
0% Patient Pool Expansion from Generics		0	0	0	0	0	
Total GLP-1 Elastomer Revenue	233	280	350	420	483	531	17.9%
Remainder of STVN (€mn)	953	991	1,031	1,072	1,115	1,159	4.0%
Remainder Growth	0.3%	4.0%	4.0%	4.0%	4.0%	4.0%	
Total (€mn)	1,186	1,271	1,381	1,492	1,598	1,691	7.3%
Scenario #1	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 (€mn)	233	280	350	420	483	531	17.9%
GLP Revenue Growth	51.7%	20.0%	25.0%	20.0%	15.0%	10.0%	
10% Patient Pool Expansion from Generics		23	28	35	42	48	
Total GLP-1 Elastomer Revenue	233	303	378	455	525	580	20.0%
Remainder of STVN (€mn)	953	991	1,031	1,072	1,115	1,159	4.0%
Remainder Growth	0.3%	4.0%	4.0%	4.0%	4.0%	4.0%	
Total (€mn)	1,186	1,294	1,409	1,527	1,640	1,739	8.0%
Scenario #2	2025	2026	2027	2028	2029	2030	CAGR (2025-2030)
Base Case Branded GLP-1 (€mn)	233	280	350	420	483	531	17.9%
GLP Revenue Growth	51.7%	20.0%	25.0%	20.0%	15.0%	10.0%	
20% Patient Pool Expansion from Generics		47	56	70	84	97	
Total GLP-1 Elastomer Revenue	233	327	406	490	567	628	21.9%
Remainder of STVN (€mn)	953	991	1,031	1,072	1,115	1,159	4.0%
Remainder Growth	0.3%	4.0%	4.0%	4.0%	4.0%	4.0%	
Total (€mn)	1,186	1,318	1,437	1,562	1,682	1,787	8.5%

Source: Morgan Stanley Research estimates

Our analysis found generic GLP-1s could drive 30-50bps of incremental growth for WST assuming generics could expand the GLP-1 patient pool by 10-20%. Using the same assumptions for GLP-1 patient pool expansion for STVN, we found generic GLP-1s could drive 70-120 bps of incremental growth. The delta is explained primarily by STVN's higher company level GLP-1 exposure (19-20%) relative to WST's elastomer exposure.

Key Stock Conclusions

Originators

Novo Nordisk (EW): We expect Novo Nordisk to be under pressure in post-LOE

markets. We expect Canada to be the bellwether for future developed market dynamics over coming years; specifically, the extent to which price and volume competition in the semaglutide franchise leads to an erosion of Novo's Wegovy/Ozempic sales. We see some scope for Novo to partially compensate through its Wegovy Pill and through innovation. On Wegovy Pill, script trends in the US YTD have been encouraging (although we are waiting to see prescriptions trends holding through the summer and escalation to highest doses) but we believe the API requirement will limit Novo's ability execute on a broad ex-US launch (unlike LLY with Foundayo). On innovation, CagriSema (GLP-1/amylin) launch in 2027 will be a first step to partially protect the franchise in the US, but we see more potential from the cagrilintide mono (amylin, potential launch in 2028), zenagamtide/amycretin subcut (GLP-1/amylin 2029-2030) and more importantly zenagamtide oral (2030), which carries the highest differentiation potential in our view. At this stage, we see less potential for Novo's triple agonists which will only come to market next decade and 5+ years after Lilly's retatrutide. Therefore, with key new launches skewed to 2028-2030+, we see a risk for Novo's diabetes/obesity franchise to struggle between inexpensive semaglutide generics and innovation from Lilly and other branded players, and we see consensus' forecasts for \$23bn sales for semaglutide in 2033 as likely at risk (MSe \$16bn).

Eli Lilly (OW): We expect Eli Lilly's franchise to be more insulated from generic

pressure given: (1) Differentiation of Mounjaro (GLP-GIP) vs. Semaglutide (GLP) based on head-to-head Ph3 data in obesity and T2D, (2) a portfolio of pipeline drugs with novel mechanisms of action such as Retatrutide (GLP-GIP-glucagon) and Eloralintide (selective amylin), and (3) a scalable oral GLP-1 pill in Foundayo. We model LLY incretin franchise sales growing from \$60bn in 2026 to \$101bn in 2035.

Generic companies, CDMOs and suppliers

Dr Reddy's (EW): Dr. Reddy's is an early entrant in generic semaglutide in Canada, India and other markets with in-house API production and partner-led pen manufacturing/fill-and-finish. It is rapidly expanding capacity (targeting >40mn pens by FY28 vs. 6–7mn in 2026) and pursuing global markets, including established GLP-1 market like Canada. The company is positioning itself as a scaled, globally focused generics player, leveraging cost leadership, integrated manufacturing, and supply chain control to gain market share.

Divi's Labs (OW): The company aims to be one of the largest players in peptides globally, and currently has the largest Solid-Phase Peptide Synthesis (SPPS) scale (i.e. multiple ~3,000L reactors) in India. Management confirmed that the company is involved in manufacturing amino acids and fragments for the API of several MNCs GLP-1 projects, including innovative drugs. We therefore see Divi's Labs as a potential beneficiary of volume expansion in the GLP-1 market through exposure to API manufacturing.

Sandoz (UW): Sandoz is expecting to launch a generic semaglutide in Canada and Brazil in

the second half of 2026, and is targeting other markets such as Mexico in 2027 ahead of the EU and US LOE in the early/mid-2030s. The company is partnering for both API and fill-and-finish for the first markets, and will make a decision on investing in API and F&F capacity based on the market dynamics in the first LOE markets. We currently model a limited contribution from GLP-1 in 2027 in our model based on the Canada and Brazil opportunity; if Sandoz emerges as a market share leader in Canada and Brazil and prices stabilise at an acceptable level, there could be upside to our numbers from GLP-1 generics, in particular if Sandoz decides to invest in F&F manufacturing to address the US and European markets. We currently expect ~US\$100-120mn sales by 2030 for Sandoz Ozempic generics in Canada (implying 15-20% share of the Ozempic generic market in Canada).

Medtide (3880.HK, EW): One of a handful of Chinese semaglutide API suppliers on the US FDA's Green List of importers, Medtide possesses ton-scale capacity. However, it is a peptides and oligos focused CDMO that serves novel drug contracts. So far, it has not announced plans to capture the semaglutide biosimilars market, as opposed to Hybio Pharma (300199.SZ, not covered) and Jiangsu Sinopep (688076.SH, not covered) which are dedicated semaglutide API suppliers.

United Labs (3933.HK, EW): One of the first filers for a semaglutide biosimilar in China. As a well-established insulin player in China, it has the sales network to push other products from its diabetes pipeline. Novo licensed the ex-China rights to United Labs' triple agonist UBT251 (GLP-1/GIP/glucagon) in 2025 ([trial progress](#)).

Hypera (OW): The opportunity for Hypera in Brazil from a generic semaglutide launch is execution-dependent, but potentially meaningful: if the company secures timely Anvisa approval and launches in the early waves of branded semaglutide copies, it could participate directly in the expansion and formalization of a market where demand already exists but is still partly priced out of the legal channel. EMS's first-mover advantage and local manufacturing footprint are relevant competitive hurdles, but the market should evolve into a multi-player branded-copy segment, where HYPE's physician relationships, marketing capabilities, retail reach and brand-building capacity could support share gains as semaglutide shifts from a premium niche into a broader mass-market therapy.

Raia Drogasil (OW): We see upside in the expansion of the addressable market in Brazil as lower semaglutide prices move GLP-1 access beyond a narrow high-income niche, alongside the migration of demand currently served by unauthorized imports and compounded alternatives into the formal pharmacy channel. While lower prices may pressure average ticket, elasticity should support higher volumes, traffic and recurring prescription demand, with RADL well positioned to capture this shift given its national footprint, digital capabilities, supplier relationships and ability to sell both branded and generic GLP-1s.

Drug Packaging

West Pharmaceutical Services (EW): We believe generic GLP-1s will be an incremental growth driver for WST and estimate it could drive 30-50 bps of annual growth through 2030. We expect minimal product trade down from branded to generic GLP-1s, enabling WST to serve the generics GLP-1s market with its high value portfolio, which should drive meaningful topline flow through to the EPS line. In addition we see the potential for ASP

upside from lesser volume discounts as generics players will not be able to receive the same level of volume discounts compared to the two main branded GLP-1 players.

Stevanato Group (EW): We see STVN benefiting from generic GLP-1s for the same aforementioned reasons as WST. However with 19-20% company exposure to GLP-1s, we expect STVN will benefit to a even greater degree.

Share prices: Novo Nordisk 284.90 DKK, Eli Lilly 1,082.20 USD, Dr Reddy's 1,274.00 INR, Divi's Labs 6,533.00 INR, Sandoz 64.22 CHF, Medtide 19.63 HKD, United Labs 9.07 HKD, Hypera 2143 BRL, Raia Drogasil 17.86 BRL, Hims & Hers 27.51 USD, WST 317.32 USD, STVN 18.51 USD.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Artur D Alves; Thibault Bouterlin; Mauricio I Cepeda; Terence C Flynn, Ph.D.; Sarita Kapila; Binay Singh; Laurence Tam; Kallum L. Titchmarsh.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of April 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AstraZeneca Plc, Bayer AG, Dr Reddy's Laboratories Ltd, Eli Lilly & Co., Galderma Group AG, Grifols SA, GSK plc, Hims & Hers Health Inc, Hypera SA, Lonza Group AG, Novo Nordisk A/S, Raia Drogasil SA, Sanofi SA, Sartorius Stedim Biotech SA, Stevanato Group S.p.A., Teva Pharmaceutical Industries Ltd., West Pharmaceutical Services Inc..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of AstraZeneca Plc, Eli Lilly & Co., Galderma Group AG, Hims & Hers Health Inc, Medtide, Sanofi SA.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from AstraZeneca Plc, Bayer AG, Eli Lilly & Co., Grifols SA, GSK plc, Hims & Hers Health Inc, Medtide, Novo Nordisk A/S, Sanofi SA.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Alvotect SA, AstraZeneca Plc, Bayer AG, Dr Reddy's Laboratories Ltd, Eli Lilly & Co., Galderma Group AG, Grifols SA, GSK plc, Hims & Hers Health Inc, Hypera SA, Lonza Group AG, Medtide, Merck KGaA, Novartis AG, Novo Nordisk A/S, Roche Holding AG, Sandoz Group AG, Sanofi SA, Sartorius AG, Stevanato Group S.p.A., Teva Pharmaceutical Industries Ltd., UCB S.A., West Pharmaceutical Services Inc..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from Alvotect SA, AstraZeneca Plc, Bayer AG, Eli Lilly & Co., Galderma Group AG, Grifols SA, GSK plc, Lonza Group AG, Merck KGaA, Novartis AG, Novo Nordisk A/S, Roche Holding AG, Sanofi SA, Stevanato Group S.p.A., Teva Pharmaceutical Industries Ltd., The United Laboratories, UCB S.A..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Alvotect SA, AstraZeneca Plc, Bayer AG, Dr Reddy's Laboratories Ltd, Eli Lilly & Co., Galderma Group AG, Grifols SA, GSK plc, Hims & Hers Health Inc, Hypera SA, Lonza Group AG, Medtide, Merck KGaA, Novartis AG, Novo Nordisk A/S, Raia Drogasil SA, Roche Holding AG, Sandoz Group AG, Sanofi SA, Sartorius AG, Stevanato Group S.p.A., Teva Pharmaceutical Industries Ltd., UCB S.A., West Pharmaceutical Services Inc..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Alvotect SA, AstraZeneca Plc, Bayer AG, Eli Lilly & Co., Galderma Group AG, Grifols SA, GSK plc, Hims & Hers Health Inc, Lonza Group AG, Medtide, Merck KGaA, Novartis AG, Novo Nordisk A/S, Roche Holding AG, Sanofi SA, Stevanato Group S.p.A., Teva Pharmaceutical Industries Ltd., The United Laboratories, UCB S.A.. Morgan Stanley & Co. LLC makes a market in the securities of Alvotect SA, Dr Reddy's Laboratories Ltd, West Pharmaceutical Services Inc..

Morgan Stanley & Co. International plc is a corporate broker to AstraZeneca Plc, Lonza Group AG.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade

as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of May 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

A member of Research who had or could have had access to the research prior to completion owns securities (or related derivatives) in the Eli Lilly & Co.. This person is not a research analyst or a member of research analyst's household.

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic

basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. The recommendations of Artur D Alves; Mauricio I Cepeda in this report reflect solely and exclusively the analyst's personal views and have been developed independently, including from the institution for which the analyst works.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group

Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 14-9169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products. Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Pharmaceuticals

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/04/2026)
Sarita Kapila		
AstraZeneca Plc (AZN.L)	O (02/12/2025)	13,554p
AstraZeneca Plc (AZN.N)	O (02/12/2025)	US\$181.80
GSK plc (GSK.L)	U (02/12/2025)	1,902p
GSK plc (GSK.N)	U (02/12/2025)	US\$51.27
Roche Holding AG (RHHBY.PK)	E (04/24/2026)	US\$51.57
Roche Holding AG (ROPC.S)	E (04/24/2026)	SFr 324.60
Sanofi SA (SASY.PA)	E (05/01/2026)	€76.44
Sanofi SA (SNY.N)	E (05/01/2026)	US\$44.39
UCB S.A. (UCB.BR)	E (03/06/2026)	€259.50

Thibault Bouterin

Alvotect SA (ALVO.O)	O (10/14/2025)	US\$3.74
Bayer AG (BAYGn.DE)	O (12/03/2025)	€35.56
Galderma Group AG (GALD.S)	E (05/02/2024)	SFr 159.15
Grifols SA (GRLS.MC)	E (03/10/2026)	€9.05
Lonza Group AG (LONN.S)	O (11/25/2025)	SFr 491.40
Merck KGaA (MRCG.DE)	E (02/12/2025)	€139.50
Novartis AG (NOVN.S)	O (10/31/2025)	SFr 115.84
Novartis AG (NVS.N)	O (10/31/2025)	US\$148.40
Novo Nordisk A/S (NOVOb.CO)	E (03/03/2026)	DKr 283.85
Novo Nordisk A/S (NVO.N)	E (03/03/2026)	US\$43.75
Sandoz Group AG (SDZ.S)	U (02/27/2026)	SFr 63.60
Sartorius AG (SATG_p.DE)	O (02/12/2025)	€249.80
Sartorius Stedim Biotech SA (STDM.PA)	O (02/12/2025)	€183.70
Teva Pharmaceutical Industries Ltd. (TEVA.N)		US\$34.43

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.